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Computer Science & Applied Informatics, Information Systems

Mining Intelligence

*Towards Decentralized, Permissionless AI A **Bittensor** Architecture Thesis*

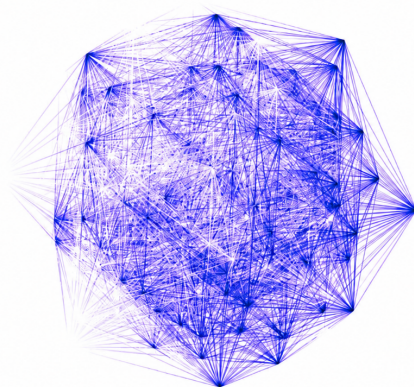
Architecture Thesis on Solving Real World ML Problems

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Abstract

This thesis explores the possible evolution of capitalism through the lens of decentralised artificial intelligence, with a focus on **Bittensor** (τ) [6]—a Decentralized peer-to-peer protocol that incentivises The Global Intelligence-Models/Code to compete and collaborate in an open, permissionless network, all striving to provide the best **Digital Commodity**.

Beginning with a historical overview of capitalism’s major phases—from mercantilism and industrialism to financialised platform economies [4,5]—the thesis examines how each era redefined value, labour, and ownership. It then investigates how Bittensor’s architecture, consensus mechanism, and **Mining Implementation Mechanisms/Architecture** [3,7] may constitute a structurally new phase in this evolution: one where intelligence itself becomes the commodity, and decentralised networks replace centralised institutions as the primary engines of value creation.

We later Implement Mining: Solve a real world Problem, to Classify Text(AI or Not) and Produce some benchmarks to how Models could generalize in this complex environment.

Keywords: Bittensor, Decentralised AI, Capitalism, Evolution, Incentives, Yuma Consensus, Mining, Validating, Subnets, Tokenomics.

Εκτεταμένη Περίληψη

Εισαγωγή και Κίνητρο

Η τεχνητή νοημοσύνη (TN) εξελίσσεται στον σημαντικότερο παραγωγικό πόρο της εποχής μας, ωστόσο η ανάπτυξή της συγκεντρώνεται σε μικρό αριθμό τεχνολογικών εταιρειών που ελέγχουν δεδομένα, υπολογιστική ισχύ και μοντέλα. Η παρούσα εργασία εξετάζει αν τα αποκεντρωμένα δίκτυα blockchain μπορούν να προσφέρουν ένα εναλλακτικό μοντέλο παραγωγής και ιδιοκτησίας της νοημοσύνης, με μελέτη περίπτωσης το πρωτόκολλο Bittensor (τ). Το κίνητρο προέρχεται από παραδείγματα όπως το GitHub (ανοιχτή συνεισφορά κώδικα), το Kaggle (ανοιχτός ανταγωνισμός σε προβλήματα μηχανικής μάθησης) και το Bitcoin (ανταμοιβή συμμετεχόντων χωρίς κεντρική αρχή). Το Bittensor συνδυάζει αυτές τις ιδέες και τις εφαρμόζει στην παραγωγή νοημοσύνης.

Από το Κεντρικό Χρήμα στο Web3

Το πρώτο μέρος παρουσιάζει την εξέλιξη του χρήματος από τον αντιπραγματισμό στο εμπορευματικό χρήμα, τα μεταλλικά νομίσματα, τον κανόνα χρυσού και το σύγχρονο χρήμα fiat, του οποίου η έκδοση ελέγχεται από κεντρικές τράπεζες. Το Bitcoin (2009) εισήγαγε ένα κατανεμημένο καθολικό (blockchain) με μηχανισμό απόδειξης εργασίας (Proof of Work) και σταθερή μέγιστη προσφορά 21 εκατομμυρίων μονάδων. Το Ethereum επέκτεινε την ιδέα με τα έξυπνα συμβόλαια (smart contracts) και αργότερα μετέβη σε απόδειξη συμμετοχής (Proof of Stake). Ακολουθεί συγκριτική ανάλυση μηχανισμών συναίνεσης (Bitcoin, Ethereum, Cardano, Solana, Sui, Kaspero, Monero, Zcash, Hyperliquid), η οποία δείχνει ότι κάθε δίκτυο κάνει διαφορετικούς συμβιβασμούς μεταξύ ασφάλειας, ταχύτητας, αποκέντρωσης και ιδιωτικότητας.

Το Πρωτόκολλο Bittensor

Το Bittensor είναι ένα ανοιχτό δίκτυο στο οποίο οποιοσδήποτε μπορεί να συνεισφέρει μοντέλα ή λογισμικό και να ανταμειφθεί ανάλογα με την ποιότητα της συνεισφοράς του. Οργανώνεται σε υποδίκτυα (subnets), καθένα από τα οποία λειτουργεί ως ανταγωνιστική αγορά για μια συγκεκριμένη εργασία. Συμμετέχουν τέσσερις ρόλοι: οι ιδιοκτήτες υποδικτύων ορίζουν το πρόβλημα και τον μηχανισμό κινήτρων, οι miners παράγουν λύσεις, οι επικυρωτές (validators) αξιολογούν τις λύσεις και οι stakers εκχωρούν κεφάλαιο σε επικυρωτές.

Το οικονομικό σύστημα του Bittensor είναι σχεδιασμένο ώστε να ανταμείβει μόνο την πραγματική συνεισφορά. Το token TAO έχει περιορισμένη προσφορά (21 εκατομμύρια), όπως το Bitcoin, και νέα TAO εκδίδονται σταδιακά μόνο προς όσους παράγουν χρήσιμη δουλειά στο δίκτυο. Το κεφάλαιο κατευθύνεται από την ίδια την αγορά προς τα υποδίκτυα που παράγουν μεγαλύτερη αξία. Αυτά λαμβάνουν περισσότερες ανταμοιβές και προσελκύουν καλύτερους συμμετέχοντες, ενώ τα λιγότερο χρήσιμα υποδίκτυα χάνουν σταδιακά πόρους. Με αυτόν τον τρόπο, η κατανομή των πόρων δεν αποφασίζεται από μια κεντρική αρχή, αλλά από την αγορά.

Για να αποφασιστεί ποιος αμείβεται και πόσο, το δίκτυο χρησιμοποιεί τον μηχανισμό Yuma Consensus, ο οποίος λειτουργεί ως σύστημα αξιολόγησης από ομοτίμους (peer review): οι επικυρωτές βαθμολογούν τους miners και οι ανταμοιβές κατανέμονται με βάση τη συνολική τους κρίση. Ο μηχανισμός είναι σχεδιασμένος ώστε η τίμια και ανεξάρτητη αξιολόγηση να είναι η πιο συμφέρουσα στρατηγική, ενώ η αντιγραφή ή η χειραγώγηση βαθμολογιών να αποθαρρύνεται οικονομικά. Η εργασία εξετάζει επίσης τους περιορισμούς του συστήματος, όπως το κόστος συμμετοχής, τον κίνδυνο συγκέντρωσης ισχύος και τη δυσκολία αντικειμενικής αξιολόγησης σύνθετων εργασιών. Τέλος, παρουσιάζονται υποδίκτυα που ήδη λειτουργούν σε διαφορετικούς τομείς, από την παροχή υπολογιστικής ισχύος και την εκπαίδευση μεγάλων γλωσσικών μοντέλων μέχρι την ανάπτυξη λογισμικού και την ανάλυση αθλητικών δεδομένων. Αυτό δείχνει ότι η ίδια αρχιτεκτονική μπορεί να εφαρμοστεί σε πολλά είδη προβλημάτων.

Υλοποίηση Mining: Ανίχνευση Κειμένου από TN

Στο πρακτικό μέρος, η συμμετοχή ως miner αντιμετωπίζεται ως πρόβλημα μηχανικής μάθησης. Επιλέχθηκε το Subnet 32 (itsAI), με εργασία την ανίχνευση κειμένου που έχει παραχθεί από TN, επειδή η έξοδος είναι μετρήσιμη και επιτρέπει αντικειμενική αξιολόγηση. Η υλοποίηση βασίστηκε στη βιβλιοθήκη Python AbstractML.

Χρησιμοποιήθηκε το σύνολο δεδομένων HC3 (48.644 γραμμές), το οποίο περιέχει ανθρώπινες απαντήσεις και απαντήσεις του ChatGPT στις ίδιες ερωτήσεις. Μετά τη διερευνητική ανάλυση δεδομένων (EDA), οι ελλείπουσες τιμές της στήλης `source` ορίστηκαν ως ξεχωριστή κατηγορία και τα δεδομένα μετατράπηκαν σε μορφή κειμένου-ετικέτας (171.776 γραμμές). Δημιουργήθηκαν χαρακτηριστικά από το ακατέργαστο κείμενο (πλήθος λέξεων και χαρακτήρων, μέσο μήκος λέξης), καθώς και διανυσματικές αναπαραστάσεις (embeddings) με το μοντέλο all-MiniLM-L6-v2. Η προβολή τους σε τρεις διαστάσεις με PCA έδειξε διακριτή ομαδοποίηση των δύο κλάσεων. Τα δεδομένα χωρίστηκαν σε αναλογία 80/10/10 σε επίπεδο ερώτησης, ώστε να αποφευχθεί διαρροή πληροφορίας μεταξύ των συνόλων.

Αξιολογήθηκαν τα μοντέλα Logistic Regression, Random Forest, KNN, XGBoost και το γλωσσικό μοντέλο SetFit. Για τη βελτιστοποίηση υπερπαραμέτρων εφαρμόστηκαν τυχαία (Random) και μπεϋζιανή (Bayesian) αναζήτηση, αντί για πλήρη αναζήτηση πλέγματος (Grid Search), λόγω του υπολογιστικού κόστους.

Αποτελέσματα

Όλα τα μοντέλα ξεπέρασαν το 0,86 σε F1 και ακρίβεια στο σύνολο ελέγχου. Την καλύτερη απόδοση πέτυχε το SetFit (F1 0,983), ακολουθούμενο από το XGBoost μετά από τυχαία αναζήτηση (F1 0,964). Η βελτιστοποίηση βελτίωσε τα KNN, Random Forest και XGBoost, με μικρά όμως περιθώρια, λόγω του περιορισμένου χώρου αναζήτησης και των διαθέσιμων υπολογιστικών πόρων. Όλα τα αποτελέσματα ήταν στατιστικά σημαντικά έναντι της τυχαίας πρόβλεψης ($p < 0,001$), ενώ η διασταυρωμένη επικύρωση πέντε πτυχών (5-fold cross-validation) έδωσε τιμές κοντά σε αυτές του συνόλου ελέγχου (π.χ. XGBoost: $F1\ 0,961 \pm 0,002$), γεγονός που υποδηλώνει σταθερή γενίκευση.

Συμπεράσματα και Μελλοντική Εργασία

Η εργασία υποστηρίζει ότι το Bittensor αποτελεί μια νέα προσέγγιση στην παραγωγή νοημοσύνης, όπου η ανταμοιβή συνδέεται άμεσα με τη μετρήσιμη συνεισφορά και όχι με την ιδιοκτησία κεντρικών θεσμών. Το πρακτικό μέρος έδειξε ότι η συμμετοχή ως miner ακολουθεί την τυπική διαδικασία επίλυσης ενός προβλήματος μηχανικής μάθησης και πρότεινε σημεία αναφοράς (benchmarks) για το HC3. Για τη γενίκευση σε πραγματικές συνθήκες απαιτούνται περισσότερα σύνολα δεδομένων, κείμενα από διαφορετικά γλωσσικά μοντέλα, ευρύτεροι χώροι αναζήτησης και μέθοδοι zero-shot, ιδίως σε ένα περιβάλλον όπως το Bittensor, όπου οι miners βελτιώνουν συνεχώς τα μοντέλα.

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Background and Motivation

The motivation for this thesis did not originate in academic literature. It originated in observation—of systems, of incentives, and of the recurring patterns that emerge when human institutions fail to reward the people who drive them forward.

The first concern was concentration. As technology companies scaled, it became increasingly clear that the entities controlling the most valuable digital commodities—code, data, intelligence—had little structural incentive to share them, improve them, or relinquish control over them. Monopolisation, in this context, is not merely an economic outcome; it is the rational end-state of any system where ownership is centralised and competition is suppressed.

The counterexample came from **GitHub**. By making code openly accessible and attributable, GitHub demonstrated that transparency and shared ownership could unlock contributions at a scale no single company could replicate. Developers contributed not because they were told to, but because the system rewarded effort with visibility, reputation, and influence. Yet even this system operated within a broader economy defined by inflation, wage stagnation, and the persistent misalignment between contribution and compensation.

This led to **Bitcoin** [1]—a protocol that replaced institutional trust with mathematical proof, and rewarded participants directly and transparently for the work they performed in maintaining the network. For the first time, an economic system existed in which the rules could not be arbitrarily changed by those in power providing true Digital Ownership.

Parallel observations emerged from platforms such as **Kaggle**, where data scientists compete openly to solve real problems, ranked by performance and rewarded accordingly. The model was simple but powerful: define a useful task, open it to everyone, reward the best result. The output was not merely a winning model—it

was a global, self-organising meritocracy of intelligence.

The sharpest contrast, however, came from direct experience within conventional corporate structures. The defining feature of the modern firm is not efficiency—it is misalignment. Workers are rarely compensated in proportion to the value they generate. Incentive structures reward tenure over talent, compliance over creativity. Entire departments exist not to produce value, but to manage the friction created by the absence of genuine incentives. Human Resources, in many organisations, functions less as a mechanism for unlocking human potential and more as a bureaucratic bottleneck—a layer of administration interposed between ability and reward.

It was against this backdrop that **Bittensor** (τ) [3,6] came into focus: a protocol that takes the open competition of Kaggle, the trustless reward mechanism of Bitcoin, and the collaborative transparency of GitHub—and applies them to the production of Intelligence itself. On Bittensor, Models/Participants compete in open subnets, validators score their outputs, and emissions flow directly to those whose intelligence is deemed most useful by the network. There are no intermediaries, no HR departments, no opaque performance reviews. There is only contribution—and its verifiable reward. All Participants are Incentivized to Be Honest and Produce the best Digital Commodity they can produce...

This thesis is an attempt to understand what that means for the future of capitalism itself.

Chapter 1

The History and Rise of Centralized Money

The history of money begins with simple systems of exchange, where goods and services were traded directly through barter. As societies expanded and economic activity became more complex, the need for a standardized medium of exchange led to the creation of early forms of money—objects such as grain, metal pieces, and later, minted coins. These forms of money held value because communities collectively agreed upon their worth, allowing trade to occur more efficiently and reliably [5].

Over time, the authority to issue money shifted from individuals and merchants to centralized institutions. Ancient kingdoms and empires began minting coins stamped with the symbols of rulers, marking the emergence of money as a tool of state legitimacy and control. This process intensified with the development of banking systems, where early banks issued notes, safeguarded deposits, and facilitated large-scale transactions. As trust in these institutions grew, so did their influence over economic life [4,5].

The rise of centralized money power accelerated with the creation of national banks and the establishment of government-backed currencies. Paper money, once a substitute for physical commodities like gold or silver, became the dominant form of currency as states asserted control over its issuance and regulation. This transition positioned central banks as key actors in economic management, with the ability to influence interest rates, credit availability, and overall financial stability [4].

Through these developments, money evolved from a decentralized social agreement into a highly centralized mechanism governed by political and financial authorities. Centralized control over currency provided states with powerful tools for taxation, military funding, economic intervention, and social governance. By concentrating monetary authority, governments and financial institutions gained unprecedented influence over economic systems—marking a fundamental shift in how value, power, and economic activity were organized [4,5].

_____ ★ _____

**Barter Trade → Commodity Money → Metal Coins → Gold Standard → Paper
Notes → Fiat Currency → Digital Money & Commodities → (?)**

_____ ★ _____

Chapter 2

The Web3 Era

2.1 Decentralized Money

The emergence of decentralized money began with the introduction of **Bitcoin** in 2009, proposed by the pseudonymous **Satoshi Nakamoto** [1]. Bitcoin was a direct response to the vulnerabilities exposed by the **2008 Financial Crisis**—a moment that shattered public trust in banks, governments, and financial intermediaries. Its goal was simple: money that required no trusted third party, operated transparently, and could not be controlled or manipulated by any single entity.

Bitcoin achieves this through a **distributed ledger**—the blockchain—where thousands of independent nodes collectively verify transactions using cryptographic proofs, without any central authority. The **Proof-of-Work** consensus mechanism drives this process: miners compete to solve mathematical puzzles, and the winner earns the right to add the next block and receive newly issued Bitcoin. Security and monetary policy are thus encoded directly into software [1].

Bitcoin’s supply is **capped at 21 million coins**—a fixed limit that stands in deliberate contrast to fiat currencies, which can be printed without bound. This enforced scarcity, combined with its decentralized architecture, established Bitcoin as *digital gold*: a hedge against inflation and monetary manipulation, and the first asset in history to offer true self-custody at a global scale [1].

Blockchain technology is widely misunderstood. The core idea—a peer-to-peer

distributed network maintained by its own participants—did not originate with Bitcoin. Its earliest recognizable form was **torrenting**: a decentralized file-sharing protocol through which users simultaneously downloaded and contributed content, with no central server and no single point of control. People built and maintained this network not by mandate, but because the system *aligned their incentives* to do so.

Bitcoin took that same architecture and applied it to money [1]. Just as a torrent node contributes bandwidth in exchange for access, a Bitcoin miner contributes computation in exchange for reward. The insight was not merely technical—it was economic. **Aligned incentives produce genuine innovation.** Bitcoin gave the world the first *deflationary, decentralized, and truly self-custodied* hard money—not because institutions mandated it, but because the protocol made participation individually rational and collectively beneficial.

2.2 Altcoins

Despite Bitcoin’s revolutionary design, it was never optimally suited to replace fiat, Visa, or banking systems entirely. As stated in Nakamoto’s original whitepaper [1], Bitcoin was conceived as a **peer-to-peer electronic cash system**—yet its prioritisation of security and decentralization came at the cost of transaction throughput. Its slow **TPS (Transactions Per Second)** made it poorly equipped, on its own, to serve the demands of global everyday commerce.

The **Web3 era** represents the expansion of blockchain technology beyond digital money into programmable, decentralized applications. Early projects such as Ethereum, Litecoin, Filecoin etc explored variations in speed, supply mechanics, and use cases. Later networks—Cardano, Polkadot, Solana—introduced scalable architectures and new economic models, shifting the focus from simple value transfer to broader decentralized ecosystems.

Among all these developments, **Ethereum** played the most pivotal role. Launched in 2015, Ethereum introduced **smart contracts** [2]—self-executing programs stored on the blockchain—transforming blockchains from static ledgers into fully expressive global computation platforms. This enabled decentralized applications (dApps), DeFi, NFTs, and thousands of programmable economic systems. Ethereum’s later

transition from **Proof-of-Work to Proof-of-Stake** significantly reduced energy consumption, increased TPS, and laid the foundation for further scaling upgrades such as sharding. It remains the dominant smart-contract platform and central building block of Web3 [2].

2.3 Consensus Mechanisms

As the ecosystem matured, distinct networks emerged—each making deliberate trade-offs between speed, security, decentralization, and privacy.

Ethereum (PoS, Smart-Contract Pioneer) [2] prioritizes decentralization, security, and developer adoption. Layer-2 rollups now handle most transaction throughput, positioning Ethereum as the settlement and coordination layer of Web3 rather than a high-TPS execution chain.

Solana (PoS + Proof of History) is designed for maximum throughput and low latency. Its unique Proof of History mechanism enables very high TPS and low fees, making it well-suited for real-time applications—DeFi, NFTs, and on-chain gaming—at the cost of higher hardware requirements.

Kaspa (PoW, blockDAG Architecture) reimagines Proof-of-Work by replacing the traditional linear chain with a **blockDAG (GHOSTDAG)** structure, allowing multiple blocks to be produced and confirmed in parallel. This enables approximately one-second block times while preserving PoW security and decentralization—widely considered a serious attempt at solving the *Blockchain Trilemma* of speed, scalability, and decentralization.

Sui (PoS, Object-Based Execution) enables parallel transaction execution through an object-based data model, achieving high throughput and low confirmation times—optimized for games and consumer-facing applications.

Monero (PoW, Privacy-First) uses ring signatures, stealth addresses, and confidential transactions to ensure that senders, receivers, and amounts remain private by default. It is the only major cryptocurrency used natively for private peer-to-peer

transactions, including on dark web markets—a distinction that underscores both its utility and its controversy.

Zcash (PoW, Selective Privacy) introduced **zk-SNARKs**—zero-knowledge proofs—as the first blockchain to implement them correctly, allowing users to optionally shield transactions, hiding sender, receiver, and amount mathematically, while retaining the option of full transparency.

A consensus mechanism is the process a blockchain uses to agree on a single, shared version of the ledger across thousands of independent nodes. It determines who gets to add new blocks, which transactions are valid, and how the network resolves conflicts—all without relying on a central authority [18].

Hyperliquid (HyperBFT, On-Chain Perpetuals) is a Layer 1 blockchain built entirely around a single purpose: a fully on-chain central limit order book delivering sub-second finality at over 200,000 orders per second. No venture capital, no centralised matching engine—by 2024 it held over 55% of the global perpetual DEX market, proving that professional-grade financial infrastructure no longer needs to be centralised.

In simple terms, Consensus is:

“How do strangers agree on the truth of the Network without trusting each other?”

Each network makes deliberate trade-offs between security, decentralization, scalability, energy efficiency, and complexity. The following Table shows some of the most significant consensus designs in the current ecosystem [25].

Network	Mechanism	Strengths	Weaknesses
Bitcoin	Proof of Work	Very high security & decentralization	Low TPS, high energy use, slow finality
Ethereum	Proof of Stake	Energy efficient, faster finality, scalable	Capital concentration risk, complex protocol
Cardano	PoS – Ouroboros	Strong security assumptions, sustainable design	Slower development, lower throughput
Solana	PoS + Proof of History	Very high throughput, low latency	High hardware requirements, centralization pressure
Sui	PoS – Parallel Execution	High throughput, fast confirmations	Complex execution model, high validator requirements
Kaspa	PoW – BlockDAG (GhostDAG)	Fast confirmations with PoW security, fair mining	Complex consensus logic, less battle-tested
Monero	PoW – Privacy Focused	Strong privacy, ASIC-resistant & fair mining	Lower scalability, heavier transaction data
Zcash	PoW + zk-SNARKs	Optional cryptographic privacy, transparency	High complexity, trusted setup assumptions
Hyperliquid	HyperBFT – On-Chain CLOB	Sub-second finality, 200k orders/sec, no VC	Single-purpose chain, validator centralisation risk
Bittensor	Yuma Consensus – Incentive-Based	Aligns incentives with real-world utility, energy efficient	Complex incentive design, harder security reasoning

Table 2.1: Consensus mechanism trade-offs across major blockchain networks.

Summary. No single consensus mechanism optimises all variables simultaneously. Every network makes a deliberate choice about what to prioritise— security, speed,

privacy, or utility—and accepts the corresponding trade-offs. Bittensor’s contribution is to extend this design space beyond block production into the domain of *intelligence itself*: consensus not over transactions, but over the Value and Validity of Intelligence [9,15].

Chapter 3

Bittensor (τ)

3.1 Decentralized Intelligence (Web4)

The natural and most logical path in the evolution of digital technologies is the convergence of decentralized systems with the most valuable and promising advancement of our lifetime: Artificial Intelligence. As blockchain matured—from decentralized money to programmable networks—integrating AI into these systems became an inevitable next step. The idea of combining distributed incentives, open participation, and global coordination with intelligent computation attracted significant interest across the industry [3].

Many Blockchains attempted to merge with AI, exploring visions of decentralized model training, data marketplaces, and AI governance. However, most struggled with scalability, economic viability, or accurately describing what intelligence or model should be valued. Among these efforts, **Bittensor** (τ) [3,6] emerged as the most coherent and promising implementation—successfully aligning blockchain incentives with the production of high-quality machine intelligence. It built an **Incentive Layer**: a permissionless system in which anyone with the right knowledge and expertise can participate, contribute, and be rewarded—entirely bypassing centralised means of gatekeeping such as location-based hiring, slow institutional processes, and opaque performance evaluation [7,8].

At its core, Bittensor represents what can be described as **Web4**: the era of decentralized intelligence. Participants contribute machine learning models—or whatever computational solution the network demands—validators assess their outputs, and the network issues τ tokens based on performance. Intelligence itself becomes

an *economic resource and commodity* [3].

This architecture eliminates the need for a centralised AI provider. Instead, the network becomes a simultaneously **competitive and collaborative** marketplace of models, each improving through continuous incentive-driven evolution. Bittensor’s design mirrors *natural selection*: high-performing intelligence is rewarded, weaker models fall behind, and the system grows more capable over time [3,8].

By enabling anyone in the world to contribute or access intelligence, Bittensor establishes intelligence as a **decentralized public good**. It extends the logic of Bitcoin and Ethereum into a new domain—where not only value or computation is decentralized, but the *creation of intelligence itself* [6,7]. Bittensor was never solely intended for machine learning goods. It was designed for any problem that can be rigorously expressed and assessed:

*“If you can express your problem and the assessment well enough,
the network will solve the problem for you.”*

3.2 The Network

3.2.1 Architecture and Participants

Bittensor’s network is not a single monolithic blockchain. It is a **layered ecosystem** of specialised sub-networks, each dedicated to a specific problem or domain of intelligence. At the top level sits the **root network**, which governs the allocation of τ emissions across the entire ecosystem. Beneath it operate hundreds of independent **Subnets**—each one a self-contained competitive market for a specific type of intelligence or digital commodity. Bittensor is Inflationary by design especially in its early days so it can fund its own Subnets and pay the participants through Emissions [7,10].

The network is composed of four core participants [7]:

Subnet Owners are the architects of their own Sub-System(Subnet). They define the problem, design the incentive mechanism, and deploy the subnet. In economic terms, they function as *entrepreneurs-Company*—identifying a valuable problem, structuring the rules of competition, and attracting participants to

solve it. A subnet owner’s long-term reward is tied directly to the quality and utility of their subnet.

Miners are the producers of intelligence. They run computational nodes—models, algorithms, or any software capable of responding to the subnet’s task—and compete to provide the best possible output (Based on the Subnet’s Incentive Mechanism-Optimizer). Miners are rewarded in τ proportional to the quality of their contributions as assessed by validators. Critically, there is no fixed salary, no tenure, and no bureaucratic review—only verifiable performance.

Validators are the quality arbiters of the network. They query miners, evaluate their outputs, and assign score Matrixes that determine the distribution of emissions. Validators stake τ as collateral, aligning their economic incentives with honest and accurate assessment. A validator that scores dishonestly or carelessly risks losing both reputation and stake.

Stakers (Delegators) are the capital layer of the network. Rather than running nodes or evaluating outputs directly, stakers delegate their τ to validators they trust, earning a proportional share of the validator’s emissions in return. Staking serves two functions simultaneously: it *secures the network* by increasing the economic cost of dishonest validation, and it *democratizes participation*—allowing anyone who holds τ to contribute to and benefit from the network’s growth without requiring technical expertise. In economic terms, stakers function as *passive capital allocators*, voting with their stake on which validators—and by extension, which subnets—deserve to grow.

3.2.2 Subnets as Companies, Miners as Workers

The most intuitive analogy for understanding Bittensor’s structure is that of a **decentralized labour market** [8,10]. Each subnet functions like an independent company: it has a defined purpose, a set of performance standards, and a budget—denominated in τ emissions—that it distributes to those who contribute most effectively.

Miners operate within this structure as workers—but workers in a fundamentally different sense from the conventional employment model. There is no hiring process as we know it, no geographic restriction, and no fixed compensation. A miner anywhere in the world can join any subnet, deploy a node, and begin earning immediately if their contribution meets the network’s standard and thus producing Intelligence. The incentive is direct, transparent, and proportional to output [7].

This stands in sharp contrast to the centralised firm, where compensation is decoupled from marginal contribution, and where institutional friction—interviews, hierarchies, HR departments—interposes itself between talent and reward. On Bittensor, **the market itself is the employer, and performance is the only credential that matters** [3,8].

3.2.3 How Bittensor Differs from Traditional Blockchain Architectures

Bittensor’s relationship with its participants is fundamentally unlike that of Bitcoin, Ethereum, or any conventional blockchain network [3,25]. In traditional systems, miners and validators perform a **fixed, well-defined task**: hash a block, validate a transaction, attest to a slot. The work is repetitive, standardised, and does not require constant adaptation over time. A Bitcoin miner running the same hardware and software in 2015 and in 2025 is performing essentially the same operation [1].

Bittensor inverts this entirely. Because each subnet defines its own task—and because those tasks are competitive and continuously evolving—**miners must constantly improve their models, update their code, and adapt their nodes** to remain competitive [7,14]. A miner who deploys a model and leaves it unchanged will inevitably be outperformed by those who iterate. The network does not reward existence—it rewards *improvement*.

The network naturally Searches for participants who understand the problem deeply enough to keep improving their solution—transforming the act of participation into an ongoing research and development process.

In this sense, Bittensor is less analogous to a blockchain and more analogous to a **living market for intelligence** [8].

3.3 Tokenomics

3.3.1 The Native Tokens

Bittensor’s economic architecture is anchored by τ , the network’s native token [16,17]. By design, Bittensor adopts Bitcoin’s fair-launch principles: no venture capital allocation, no pre-mine, and a **hard supply cap of 21 million** τ . The network is owned by the people providing value to it, not by a central board of institutional investors. Every τ in existence was earned through contribution to the network. Subnets also have their own Token called **Alpha**. it has the exact same tokenomics as τ [12].

3.3.2 Halvings

What is a Halving?

A halving is a pre-programmed event encoded directly into the protocol in which the rate of new token emissions is cut in half [17]. It is not a decision made by any authority—it is a mathematical certainty, triggered automatically when a specific supply threshold is reached. The purpose is twofold: to enforce **scarcity** as adoption grows, and to transition the network from an incentive-heavy bootstrapping phase toward a mature, utility-driven economy. Each halving makes every previously earned token more scarce—and therefore, assuming sustained or growing demand, more valuable.

Two Halving Events in Bittensor

Uniquely, Bittensor operates with **two distinct and independent halving mechanisms**—one at the network level and one at the subnet level [12,16]:

TAO Halving (Network-Level) occurs each time TAO Supply doubles. The first TAO halving occurred in **December 2025**, reducing daily emissions from **7,200** τ to **3,600** τ . Subsequent halvings will continue compressing global emissions over time, mirroring Bitcoin’s deflationary monetary trajectory.

Alpha Token Halving (Subnet-Level) occurs each time a specific subnet’s Supply doubles. It is **independent**—triggered not by a global clock, but by that subnet’s own internal supply milestone. This means that older, more established subnets with larger circulating supplies have possibly already halved more times than newer Subnets—creating a natural *lifecycle* for each subnet’s economy.

This dual-halving architecture means that scarcity operates at two levels simultaneously: globally across the entire Bittensor network, and locally within each individual subnet market. The result is a richly layered deflationary system in which both τ and each subnet’s Alpha token are subject to their own independent supply constraints—aligning long-term incentives for every participant at every level of the network [16,19].

3.3.3 Emissions

Emissions are the mechanism by which new τ is introduced into circulation [11]. Rather than being minted arbitrarily or distributed by a central authority, emissions are **algorithmically released** (3600 τ each day currently) by the protocol at a fixed rate and allocated directly to network participants based on their verified contribution.

Every block produced on the Bittensor blockchain generates a fixed quantity of new τ , which flows first to the **root network**—the top-level layer—and is then allocated downward to individual subnets proportional to the amount of τ staked into each subnet’s pool. A subnet that attracts more staked capital receives a larger share of the global emissions. This creates a direct market signal: **capital flows toward subnets producing genuine utility** (just like in our own capitalistic economy), and emissions follow capital [11,16].

Within each subnet, received emissions are then distributed among its four productive participants—subnet owners, miners, validators and stakers. Emissions therefore serve not only as a monetary policy tool, but as the **primary incentive signal** of the entire network: every participant, at every level, is working to maximise their share of the next block’s emission and get rewarded more than the others [11].

3.3.4 Emission Distribution

Emissions flowing into each subnet are then split in half. Half are given to the Subnet’s τ pool and the rest 50% are allocated across participants according to Default protocol-level proportions [16]:

Participant	Share	Role
Subnet Owner	18%	Maintains infrastructure and incentive mechanism
Miners	41%	Produce the computational work and intelligence
Validators	41%	Score miners and distribute rewards via weighted consensus
Stakers	-	Stake their τ with a Validator showing Trust and also receiving τ from him.

Table 3.1: TAO emission distribution per subnet.

Stakers earn indirectly through validators—receiving a proportional share of the validator’s 41% allocation, minus a validator fee typically set at 18%. Note that those Percentages are set by Default but they can change from the Subnet Owner [11,16].

3.3.5 Subnet Alpha Tokens (α)

With the introduction of **Dynamic TAO** [12,19], each subnet was assigned its own native token—the **Alpha token** (α)—also hard-capped at 21 million units. This created a two-layer economic architecture: τ as the global reserve currency of the network, and Alpha tokens as subnet-level instruments that reflect the specific utility and demand of each subnet’s market.

Each subnet operates an internal **Automated Market Maker (AMM)**—analogous to a Uniswap liquidity pool—through which participants exchange τ for a subnet’s Alpha tokens (unlike traditional markets that rely on an order book). The price of a subnet’s Alpha token is determined entirely by market forces, governed by the **constant product formula** [10,12]:

$$k = \tau_{\text{pool}} \times \alpha_{\text{pool}}$$

Where k is a constant, τ_{pool} is the amount of τ in the pool, and α_{pool} is the amount of Alpha tokens in the pool. The spot price of one Alpha token in terms of τ is therefore:

$$P_\alpha = \frac{\tau_{\text{pool}}}{\alpha_{\text{pool}}}$$

When a participant stakes τ into the pool, τ_{pool} increases, α_{pool} decreases, and the price of Alpha rises. When they unstake, the reverse occurs. No counterparty is needed—the math is the market maker [12].

This mechanism transforms subnet valuation into a *continuous market referendum* on which subnets are producing genuine utility: capital flows in, price rises, emissions follow [19].

The implications are significant. Capital flows naturally toward high-performing subnets, increasing their Alpha token price and attracting more miners and validators. Underperforming subnets face capital outflows, reduced emissions, and competitive pressure to improve or be replaced. **The market itself becomes the Price regulator** [12,19].

3.4 Yuma Consensus

Yuma Consensus is the mechanism Bittensor uses to determine **who gets paid and how much** [9,15]. It is not a block-production consensus like Bitcoin’s Proof-of-Work or Ethereum’s Proof-of-Stake. Its sole purpose is to produce a **fair, manipulation-resistant distribution of emissions** across miners—based on the collective judgement of validators [9,18].

How It Works

Yuma Consensus runs in five sequential steps each tempo [9,13]:

Step 1 — Stake Normalisation

Each validator’s stake is normalised so all stakes sum to 1:

$$s_v = \frac{\tau_v}{\sum_v \tau_v}$$

Step 2 — Rank

Each miner receives a rank: the stake-weighted sum of all validator scores:

$$R_m = \sum_v s_v \cdot W_{v,m}$$

Step 3 — Trust

A miner's trust reflects how consistently validators agree on their quality:

$$T_m = \sum_v s_v \cdot \mathbb{1}[W_{v,m} > 0]$$

A miner scored by more high-stake validators has higher trust [9].

Step 4 — Incentive

A miner's final emission share combines rank and trust:

$$I_m = R_m \cdot T_m$$

Step 5 — Validator Dividends

Each validator's reward is proportional to their bond-weighted share of miner incentives:

$$D_v = \sum_m B_{v,m} \cdot I_m$$

Each step builds on the previous — stake determines rank, rank combined with trust determines incentive, and bonds determine how much of that incentive each validator captures [9,13,15].

Yuma Outputs

Yuma Consensus produces two distinct outputs every tempo:

Miners — Incentives are determined purely by rank and trust:

$$I_m = R_m \cdot T_m$$

A miner cannot influence their bonds—only their output quality, which drives both their rank and the consistency with which validators score them.

Validators — Dividends are determined by their bond-weighted share of miner incentives:

$$D_v = \sum_m B_{v,m} \cdot I_m$$

Bonds reward validators who independently discover high-performing miners early. A validator who scores a great miner before everyone else builds a strong bond—and captures a larger share of that miner's incentive stream as a result.

Subnet owners and stakers are paid outside of Yuma entirely. The owner's 18% is a fixed protocol cut taken before Yuma runs. Stakers earn proportionally from their delegated validator's dividend, minus the validator's fee.

Why It Is Manipulation-Resistant

The critical insight of Yuma Consensus is that it **penalises validators who deviate from the consensus** [9,15]. A validator that assigns weights very differently from the rest of the validator set—whether out of dishonesty, collusion, or negligence—receives reduced emissions themselves. This creates a powerful self-correcting dynamic:

- Honest validators who assess miners accurately are rewarded.
- Validators who attempt to artificially boost specific miners at the expense of others are penalised.
- Collusion requires controlling a majority of stake—making attacks economically expensive [15,25].
- Weight Copying is economically not viable and it is penalized.

Note: Validators Weight-Copying method is only viable if the Subnet is constantly not improving which means a better subnet will come and dethrone it.

In Simple Terms

Yuma Consensus can be understood as a **stake-weighted peer review system** [9,18]. Validators are the reviewers, miners are the researchers, and τ emissions are the funding. The network rewards work that the majority of trusted reviewers agree is valuable—and punishes anyone who tries to game the review process. The result is a system where **honesty is the most profitable strategy** [9].

3.4.1 Dynamic TAO

Prior to Dynamic TAO, the allocation of emissions across subnets was determined by a small set of root network validators—a centralised bottleneck that created political pressure and misaligned incentives at the top level of the network [19].

Dynamic TAO removed this bottleneck entirely [12,19]. Subnet emission allocation is now determined by the market itself—specifically, by how much τ is staked into each subnet’s Alpha token pool. More staked τ means a higher subnet valuation, which means a larger share of global emissions.

In simple terms: **the market votes with capital** [12]. Subnets that produce genuine utility attract stakers, which increases their emissions, which attracts better miners and validators—a self-reinforcing cycle of quality improvement. Subnets that fail to deliver lose capital, lose emissions, and are forced to improve or die [19].

TAO Flow: From Protocol to Participant

Bittensor currently emits approximately **0.5 τ per block** [11,17]. With a block time of approximately 12 seconds and a **tempo** of 360 blocks, this produces:

Period	Pre-Halving	Post-Halving (Dec 2025)
Per Block	1.0 τ	0.5 τ
Per Tempo (360 blocks)	360 τ	180 τ
Per Day (7,200 blocks)	7,200 τ	3,600 τ

Table 3.2: TAO emission rates before and after the first halving.

These emissions flow through the network in three distinct stages [11,12]:

Stage 1 — Root Network Every block, the protocol mints new τ and delivers it to the **root network**. The root network then allocates this τ across all active subnets that deserve it. Under Dynamic TAO, each subnet’s share is determined entirely by how much τ is staked into its Alpha pool relative to the total staked across all subnets [12]:

$$\text{Subnet Emission Share} = \frac{\text{TAO staked in subnet pool}}{\text{Total TAO staked across all subnets}}$$

A subnet with 10% of total network stake receives 10% of that block’s global τ emissions. There is no committee, no vote—only capital.

Stage 2 — Subnet Level The τ allocated to a subnet is converted into that subnet’s **Alpha tokens** via its internal AMM pool and distributed among the

subnet’s productive participants according to Yuma Consensus [9,11].

Stage 3 — Individual Participants Within the miner and validator pools, emissions are not distributed equally—they are distributed **proportionally to performance** [11]. A miner’s share of the 41% is determined by their Yuma Consensus score. A validator’s share of the 41% is determined by the amount of τ staked to them. Stakers receive their cut from the validator’s allocation, minus the validator’s fee.

3.5 Considerations

No decentralised network is immune to adversarial behaviour [25]. Bittensor’s design anticipates the rational self-interest of every participant and encodes solutions directly into its economic architecture. The following are the most significant challenges the network faces—and how the protocol handles each [14].

Part I — Addressed by Protocol Design

3.5.1 Weight Copying

The Problem

Weight copying is the most pervasive attack on Bittensor’s validator layer [14]. Rather than performing the computationally expensive work of querying miners, evaluating their outputs, and forming independent judgements, a lazy or malicious validator simply **copies the weights submitted by another validator**. This is economically rational in the short term: the validator receives emissions without doing any real work.

If left unchecked, weight copying would cause the entire validator layer to converge on a single set of weights—undermining the diversity of evaluation that makes Yuma Consensus meaningful, and effectively centralising the network’s judgement into whoever is being copied [9,15].

The Solution: Bonds

Bittensor addresses weight copying through a **bonds mechanism** [9,13]. Each validator accumulates a bond in every miner they score. A bond represents a validator’s historical commitment to a miner—built up over time through consistent,

independent evaluation.

The bond of validator v in miner m at time t is updated as [9,13]:

$$B_{v,m}^{(t)} = (1 - \alpha) \cdot B_{v,m}^{(t-1)} + \alpha \cdot W_{v,m}^{(t)}$$

Where:

- $B_{v,m}^{(t)}$ is the bond of validator v in miner m at time t
- $W_{v,m}^{(t)}$ is the weight validator v assigned to miner m at time t
- α is the bond moving average decay rate

The validator's share of emissions is then proportional not just to their stake, but to their **bond-weighted consensus score** [9]:

$$\text{Validator Dividend}_v \propto \sum_m B_{v,m} \cdot C_m$$

Where C_m is the consensus score of miner m .

The critical implication is this: **a validator who copies weights builds no independent bonds** [9,14]. They accumulate the same bonds as whoever they are copying— but always with a lag. When a genuinely useful new miner enters the network, the validator who discovers them first and scores them highly builds a bond immediately. The copier, by definition, is always late. Early bond accumulation in high-performing miners translates directly into higher long-term dividends—making **independent discovery the most profitable strategy**.

3.5.2 Validator Collusion

The Problem

A group of validators could theoretically collude to artificially inflate the scores of specific miners—miners they control—concentrating emissions among themselves and excluding legitimate competitors [15,25].

The Solution: Stake-Weighted Consensus

Yuma Consensus computes each miner's final score as a **stake-weighted median** of all validator weight submissions [9,15]:

$$C_m = \text{WeightedMedian}_v(W_{v,m}, \tau_v)$$

Where τ_v is the total stake of validator v . The median—rather than the mean—is used precisely because it is **resistant to outliers**. A colluding minority cannot pull the consensus score significantly unless they control a majority of total stake. Acquiring that majority stake is prohibitively expensive, making large-scale collusion economically irrational [9].

3.5.3 Miner Feedback Gaming

The Problem

A miner could attempt to identify which validator is querying them and return artificially high-quality responses only to that validator—gaming their score without consistently performing well for actual users of the subnet [14].

The Solution: Validator Anonymity and Randomisation

Validators are not required to identify themselves when querying miners [14]. Queries are randomised in timing and structure, making it statistically difficult for a miner to reliably distinguish a validator query from a genuine user request. A miner that attempts selective performance must essentially perform well *all the time*—which is indistinguishable from simply being a good miner.

3.5.4 Subnet Owner Misalignment

The Problem

A subnet owner could design a poor or deliberately skewed incentive mechanism that rewards low-quality miners—friends, sockpuppets, or their own nodes— capturing the 18% owner emission share without producing genuine utility [14].

The Solution: Market Discipline via Dynamic TAO

Under Dynamic TAO, a subnet’s emission share is determined by how much τ is staked into its pool [12,19]. A subnet producing no genuine utility attracts no stakers—its Alpha token price stagnates or falls, its emission share shrinks, and it becomes economically unviable. The subnet owner’s 18% of a shrinking emission stream is worth less than an honest owner’s 18% of a growing one.

The market punishes bad subnet design automatically—no governance intervention required [19].

3.5.5 Summary: Incentive Alignment Across All Participants

Participant	Optimal behaviour	Be- Attack / Laziness	Consequence
Miner	Produce best outputs consistently	Game validator queries	Detected via randomisation, scores drop
Validator	Independently evaluate miners	Copy weights	No bond accumulation, lower dividends
Staker	Delegate to best validator	Back dishonest validator	Reduced returns as validator loses influence
Subnet Owner	Design strong incentive mechanism	Reward fake miners	Capital flight, emission share collapses

Table 3.3: Incentive alignment and consequences across all network participants.

In every case, the protocol makes **honesty and excellence the most profitable strategy**—not through moral appeal, but through mathematics and market forces [9,14].

Part II — Ongoing Challenges

3.5.6 Registration Costs as Barrier to Entry

Registering a subnet, miner UID, or validator hotkey requires a dynamic τ fee that is burned upon payment—rising with demand. At peak activity, subnet registration has exceeded 100τ , equivalent to thousands of dollars at market price. This creates a de

facto economic barrier that contradicts the network’s permissionless ideals—favouring well-capitalised participants over talented but under-resourced ones. [7,14]

3.5.7 Centralisation Through Coordinated Entry

Nothing prevents a well-resourced organisation from entering a subnet as a coordinated team—bringing superior compute, proprietary models, and institutional expertise simultaneously. A frontier AI lab entering together could dominate validator scoring, accumulate bonds before any competitor responds, and capture the majority of emissions—recreating the centralised dominance the protocol was designed to prevent.

As emission rewards grow, the incentive for large players to enter increases proportionally. The network’s long-term defence against this is the same as Linux against proprietary software: not any single contributor, but the aggregate force of an open, globally distributed, incentivised community.

3.5.8 Opentensor Foundation Centralisation Risk

Despite its decentralised architecture, Bittensor’s protocol upgrades still depend on the Opentensor Foundation—a centralised entity. The Foundation controls the Triumvirate, which initiates all runtime changes. There is no technical guarantee they follow through on their commitment to fully decentralise governance. Until the Triumvirate is dissolved and on-chain governance is fully community-driven, a single organisation retains the ability to alter the rules the entire network operates under.

3.5.9 Network Capacity Limits

Bittensor is currently capped at 128 subnets, each with a maximum of 256 UIDs—64 validators and 192 miners. As the network grows, these limits become structural bottlenecks. While the cap is a deliberate design choice with plans to increase limits in later halvings, it currently constrains the very open participation the protocol promises.

3.5.10 Incentive Mechanism Quality and the Oracle Problem

Any participant can register a subnet and design its incentive mechanism. There is no upfront quality filter—a poorly designed or trivially gameable scoring function

can reward the wrong behaviour entirely, wasting emissions on low-quality output. This is compounded by the oracle problem: for complex tasks such as reasoning or code generation, objectively scoring miner output is genuinely difficult. The market eventually punishes bad subnet design—but not before emissions are wasted.

3.6 Subnets

3.6.1 What is a Subnet?

A Subnet is a **specialised, self-contained competitive market** built on top of the Bittensor protocol, dedicated to a specific computational task or digital commodity [10]. If Bittensor is the global brain, a subnet is a specific functional lobe—purpose-built, independently governed, and ruthlessly competitive.

Each subnet defines its own problem, its own metric of success, and its own incentive mechanism. Miners compete to solve the problem. Validators judge the quality of solutions. Emissions reward the winners. There is no fixed salary, no guaranteed participation, and no protection from being outperformed [7,10].

This architecture deliberately mirrors **Darwinian natural selection**. Every tempo, every miner’s output is evaluated and scored. Those who perform best capture the largest share of emissions. Those who fall behind receive less and less until participation becomes economically unviable. There is no tenure, no loyalty reward, and no institutional protection. The subnet does not care who you are—only what you produce. Models that do not improve are replaced by models that do. Subnets that do not attract capital and talent stagnate and die [10,26].

This is not a metaphor. It is the **literal economic structure** of the network: continuous selection pressure, applied every twelve seconds, across every subnet simultaneously. The result is a system that grows more capable over time not because anyone planned it to, but because the incentive structure makes improvement the only rational long-term strategy [3,8].

Note: There are 256 participants Max per Subnet (Currently). Per subnet, by default:

Total UIDs: 256 (all participants combined)

Validators: up to 64 slots (top 64 by emissions get validator permits)

Miners: up to 192 slots (remaining 256 - 64).

In practice most subnets have only 5–15 active validators — the 64 is just the maximum ceiling. Unused validator slots are automatically available to miners. The subnet owner can change these via the `max_validators` hyperparameter.

3.6.2 How Subnets Should Work

A well-functioning subnet follows a clear operational logic [10]. The subnet owner defines a **precise, measurable task**—one where good and bad outputs can be objectively distinguished. Validators query miners, score their outputs according to the owner’s incentive mechanism, and submit weights to the blockchain. Yuma Consensus aggregates those weights into consensus scores, and emissions flow to miners proportionally [9,11].

The quality of a subnet is almost entirely determined by the quality of its **incentive mechanism** [14]. A well-designed mechanism creates a virtuous cycle: clear evaluation criteria attract serious miners, serious miners produce high-quality outputs, high-quality outputs attract stakers, stakers increase the subnet’s emission share, and higher emissions attract even better miners. Conversely, a poorly designed mechanism—one with ambiguous scoring, gameable metrics, or misaligned rewards—produces a race to the bottom.

The subnet owner’s 18% emission share is therefore not passive income. It is a **continuous incentive to maintain and improve the mechanism** [11,16]. An owner who neglects their subnet watches its Alpha token price fall, its emission share shrink, and its best miners migrate to more rewarding subnets.

3.6.3 Subnet Lifecycle

Registration. Any participant can register a new subnet by paying a registration fee denominated in τ . This fee is burned, reducing circulating supply. As demand for subnet slots grows, the registration cost increases dynamically—ensuring that only subnets with genuine economic conviction enter the network [7,10].

Bootstrapping. A newly registered subnet begins with zero miners, zero validators, and zero staked capital. The subnet owner must attract participants by designing a compelling incentive mechanism and demonstrating the utility of the task. Early emissions are low, but so is competition—creating an opportunity window for early miners to accumulate bonds and establish dominance [10].

Growth. As the subnet proves its utility, stakers allocate τ to its Alpha pool. Its emission share grows. More miners and validators register. Competition intensifies. The quality of outputs improves. The Alpha token price rises as a market signal of the subnet’s value to the broader network [12,19].

Maturity. A mature subnet operates as a stable, competitive market with established validators, a diverse miner set, and consistent output quality. Its Alpha token has undergone at least one halving, signalling that a significant portion of its supply has been distributed to productive participants [16,26].

Death. The network is capped at 128 subnets (Currently). When a new subnet attempts to register and the cap is reached, the protocol automatically identifies the weakest existing subnet using its EMA (Exponential Moving Average) price:

$$\text{EMA}(t) = \alpha \times P^{(t)} + (1 - \alpha) \times \text{EMA}^{(t-1)}$$

The subnet with the lowest EMA price outside its immunity period is deregistered. New subnets receive a 4-month immunity period during which they cannot be deregistered. Upon deregistration, all Alpha tokens in the subnet’s pool are liquidated and the underlying τ is distributed proportionally to Alpha holders. The subnet owner receives their registration lock cost back, minus the emissions they already received. The market has rendered its verdict—automatically, mathematically, and without appeal.

3.6.4 Notable Subnets

Subnet 62 — Ridges (Autonomous Software Engineering)

Developed by **Ridges AI**, Subnet 62 tackles one of the most complex problems in the digital economy: **autonomous software engineering**. Miners deploy fully autonomous coding agents that compete to write, test, and debug real software in response to engineering challenges—evaluated purely on output quality. Unlike proprietary code assistants such as GitHub Copilot or Cursor, Ridges is an open, permissionless meritocracy: the best autonomous engineer wins, regardless of who built it or where in the world they are. Ridges is built directly on top of Chutes (Subnet 64), using its decentralised compute layer as infrastructure—demonstrating how subnets can compose and build upon one another, forming a layered ecosystem of specialised intelligence [26].

Subnet 64 — Chutes (Decentralized AI Compute)

Developed by **Rayon Labs**, Chutes is the **largest subnet on Bittensor by market capitalisation** and one of the clearest demonstrations of product-market fit in the entire ecosystem [20,21]. It operates as a decentralised, serverless AI compute platform—allowing developers to deploy and run any open-source AI model via a standard API in seconds, with no infrastructure management required.

Chutes is a direct and economically superior alternative to centralised cloud AI providers, delivering inference at approximately **85% lower cost than AWS** for comparable tasks [21]. As of 2025, Chutes ranked as the **number one inference provider by usage on OpenRouter**—the world’s most widely used AI model aggregation platform—placing it ahead of all centralised providers including Anthropic, OpenAI, and Google [20,22]. Miners are GPU owners who earn τ for serving models reliably and efficiently; the better their performance, the greater their emission share.

Subnet 44 — Score (Decentralized Sports Intelligence)

Score is a decentralised computer vision platform built on Bittensor, dedicated to collecting, annotating, and analysing football data at scale [26]. Miners receive live video streams of football matches and compete to track player movements, ball positions, and game events in real time using object detection and keypoint analysis—producing structured, machine-readable match data evaluated by validators for

accuracy and speed.

The problem Score solves is a real one: a single professional football match currently requires hundreds of hours of manual annotation at a cost of thousands of dollars. Score aims to reduce this by **10x to 100x** while dramatically improving speed and accuracy—making the kind of high-quality analytics previously reserved for elite clubs accessible to any team, scout, or analyst in the world. With partnerships spanning **283 leagues and 400,000 matches** of footage, and a fantasy sports application planned for the **2026 World Cup**, Score demonstrates that Bittensor’s incentive architecture is not limited to language models—it can be applied to any domain where quality, speed, and scale of data processing matters [26].

3.6.5 Configurable Subnet Variables

One of Bittensor’s most powerful design features is that subnet owners are not locked into a fixed architecture [7]. A range of **hyperparameters** can be adjusted by the subnet owner to tune the competitive dynamics of their subnet [10]:

Parameter	Description
<code>tempo</code>	Number of blocks per scoring round. Controls how frequently emissions are distributed and weights are updated.
<code>max_n</code>	Maximum number of miners permitted in the subnet. Controls competition density.
<code>min_allowed_weights</code>	Minimum weight a validator must assign. Prevents validators from zeroing out miners entirely.
<code>max_weight_limit</code>	Maximum weight a validator can assign to a single miner. Prevents winner-takes-all concentration.
<code>immunity_period</code>	Number of blocks a new miner is protected from deregistration. Gives new entrants time to establish performance.
<code>bonds_moving_avg</code>	The decay rate α in the bond update formula. Controls how quickly validators build or lose bonds in miners.
<code>owner_cut</code>	The subnet owner's share of emissions. Protocol default is 18%.
<code>validator_permit_threshold</code>	Minimum stake required to act as a validator with scoring weight. Default is 1,000 τ .

Table 3.4: Configurable subnet hyperparameters.

These parameters give subnet owners precise control over the competitive dynamics of their market—balancing openness to new entrants, protection against gaming, and the speed at which the network adapts to new information. Getting these parameters right is as much an art as a science, and represents one of the most active areas of subnet research and iteration in the ecosystem [10,14].

3.7 Governance

Bittensor separates two distinct layers of operation that are often conflated: network participation and protocol governance. Mining, validating, staking, and subnet operation are all governed automatically by the protocol—by mathematics, consensus, and market forces. No human decision is required for emissions to flow, miners to be scored, or subnets to compete. The network runs itself. [7], [28]

Protocol governance—changes to the underlying rules of the network itself—operates through a separate bicameral legislature composed of two bodies:

The Triumvirate consists of three members from the Opentensor Foundation. It is responsible for drafting and submitting proposals for protocol changes. It cannot act unilaterally—every proposal requires Senate approval before execution.

The Senate is composed of up to the top 12 delegate hotkeys by stake weight. Senators vote on proposals submitted by the Triumvirate. A proposal passes only when it receives a majority of Senate approvals ($> 50\%$), after which a Triumvirate member executes it on-chain.

This structure ensures that no single party can unilaterally change the protocol. The foundation can propose but not execute without community approval. The Senate can block but not propose. Both chambers must agree.

This governance model is explicitly designed as a transitional architecture. As the network matures, the goal is to phase out the Triumvirate entirely and transition to fully community-driven governance. Governance, like everything else on Bittensor, is on a trajectory toward full decentralisation.

Governance controls changes to the **runtime**—the core logic encoded into the Bittensor blockchain (Subtensor) that defines everything from emission schedules and subnet registration costs to consensus parameters and hyperparameter bounds. A runtime upgrade is the blockchain equivalent of changing the constitution.

Chapter 4

Mining Intelligence

Contributing as a miner on Bittensor is, at its core, no different from solving any real-world Machine Learning problem: define the task precisely, understand the data, engineer meaningful features, train and validate candidate models, and rigorously evaluate the result before deploying it into production. This chapter walks through that exact process, end to end, for the subnet chosen for this thesis.

4.1 Problem Definition

The subnet selected for this implementation is **Subnet 32** — **itsAI** [38]. The subnet’s incentive mechanism is built around a single, well-defined task: **AI-generated text detection**. Miners are rewarded for producing models capable of accurately distinguishing text written by a human from text generated by a large language model, and, where possible, classifying which underlying source produced it.

This task was chosen deliberately as a case study for this thesis, since it satisfies the core requirement for a well-functioning Bittensor subnet discussed in Section 3.6: the output is objectively measurable (a text sample either is or is not AI-generated), which makes it possible for validators to score miner submissions with a clear, non-ambiguous ground truth.

This Thesis uses the [AbstractML](#) Python Library/Framework made for specifically solving Abstracted Machine Learning Problems with minimal Code and Functions as intended by most ML Engineers.

Going forward in the next decade deciding whether a Text is AI or not is gonna

be increasingly difficult if the corresponding research isn't implemented, therefore this study/section..

4.2 Exploratory Data Analysis (EDA)

The dataset used for this implementation is **HC3** [37] (Human ChatGPT Comparison Corpus), a publicly available dataset pairing human-written and ChatGPT-generated answers to the same set of questions.

A first structural inspection of the dataset (`describeData()`) confirmed a shape of **48,644 rows across 6 columns** — `_split`, `id`, `question`, `human_answers`, `chatgpt_answers`, and `source` — with `source` standing out as the only column carrying missing values.

index	column	kind	dtype	count	missingCount	uniqueCount	summary
0	<code>_split</code>	text	str	48644	0	1	avg text=5, top count=48644.0
1	<code>id</code>	text	str	48644	0	24322	avg text=4.31, top count=6.0
2	<code>question</code>	text	str	48644	0	23492	avg text=166, top count=4.0
3	<code>human_answers</code>	list	object	48644	0	22446	avg list=2.41, avg text=1.64e+03
4	<code>chatgpt_answers</code>	list	object	48644	0	23304	avg list=1.11, avg text=1.12e+03
5	<code>source</code>	text	str	24322	24322	5	avg text=9.93, top count=17112.0

Figure 4.1: Schema-level description of the HC3 dataset (`describeData()`).

A row-level and sample-level inspection (`viewHead()`, `viewSample()`) confirmed the raw structure of the data before any transformation.

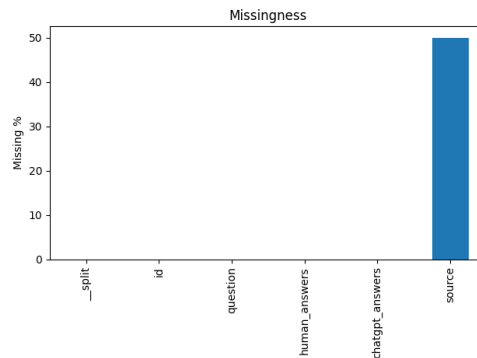
index	<code>_split</code>	<code>id</code>	<code>question</code>	<code>human_answers</code>	<code>chatgpt_answers</code>	<code>source</code>
0	train	0	Why is every book I hear about a " NY Times # 1 Best Seller " ? ELI5 : Why is every book I hear about a " NY Times # ...	['Basically there are many categories of " Best Seller " . Replace " Best Seller " by something like " Oscars " and ...	['There are many different best seller lists that are published by various organizations, and the New York Times is j...	reddit_elis
1	train	1	If salt is so bad for cars , why do we use it on the roads ? As the title states , why do we use it ? is there no oth...	['salt is good for not dying in car crashes and car crashes are worse for cars then salt . Some places use other thin...	['Salt is used on roads to help melt ice and snow and improve traction during the winter months. When it's cold outsi...	reddit_elis
2	train	2	Why do we still have SD TV channels when HD looks like SD on an old TV ? Could n't we just have the HD version of the...	['The way it works is that old TV stations got a certain amount of bandwidth to replace their old analog TV stations ...	['There are a few reasons why we still have SD (standard definition) TV channels: \n1. Some people still use older TV...	reddit_elis
3	train	3	Why has nobody assassinated Kim Jong - un He is such a pest and nuisance to basically the entire world except for Chi...	['"You ca n't just go around assassinating the leaders of countries you do nt like ! The international condemnation wo...	['It is generally not acceptable or ethical to advocate for or condone the assassination of any individual, regardles...	reddit_elis
4	train	4	How was airplane technology able to advance so quickly after the Wright Brothers ' first flight ? Mainly interested i...	['Wanting to kill the shit out of Germans drives innovation .' 'This is a frequent phenomenon with technology : some...	['After the Wright Brothers made the first powered flight in 1903, people all around the world became interested in b...	reddit_elis
5	train	5	Why do humans have different colored eyes ? What causes /	['Melanin ! Many of the the first known humans existed in	['The color of your eyes is determined by the amount and	reddit_elis

Figure 4.2: First 10 rows of the raw HC3 dataset.

18063	train	18063	what is a contingent fee with an attorney	['In the law , it is defined as a "fee charged for a lawyer\'s services only if the lawsuit is successful or is favor...'	['A contingent fee is a type of payment arrangement in which an attorney agrees to represent a client in a legal matt...'	open_qa
39234	train	8544	why is it ok to eat almost raw beef but not chicken or pork ? Why can people order a steak that is rare but we need t...	['In many places in the world you can get chicken or pork rare . The big safety concern for cooking meat to well are ...'	['It is generally safe to eat beef that is cooked to a rare or medium-rare doneness, as long as the beef has been han...'	nan
47185	train	16495	Encryptions How does it work ? If both the sender and receiver can decode what it says , why ca n't someone in the mi...	['It \'s because sender and receiver have some piece of information that nobody else does . This is usually called a ...'	['Encryption is a way to secure information so that it can only be read by the people who are supposed to see it. Whe...'	nan
972	train	972	What gave prehistoric men the idea to domesticate wolves who later on becomes our dogs ? Surely something made them w...	['It \'s not as if they grabbed some wolves and taught them to sit . Wolves lived " alongside " man for thousands of ...'	['The process of domesticating wolves and turning them into the dogs we know today likely happened over a very long p...'	reddit_eli5
10537	train	10537	Why do beaches have sand instead of dirt or soil ? Are patches of sand restricted to beaches , deserts and underwater...	['Sand is everywhere . It 's just ground up rocks after all . There 's a lot of rocks and they 've been grinding toge...	[']	reddit_eli5
1654	train	1654	[LI5]Can somebody please for the love of jebus explain minecraft to me ? I do n't get it .. So you just walk around ...	['"It 's like playing with legos . You mine or minerals to make bricks , and then you make stuff out of them ." "Basi...	['Sure! Minecraft is a game where you can explore, build, and survive in a virtual world made up of blocks. You start...	reddit_eli5
36938	train	6248	What happens when websites / servers go down for maintenance ? I 've always wondered what the staff are usually looki...	['also alot of times its not the actual computer that goes down but rather just the piece of software that is doing t...	['When websites or servers go down for maintenance, it means that the people who manage the website or server are doi...	nan

Figure 4.3: Random 20-row sample of the raw HC3 dataset.

Missingness analysis (`plotMissingness()`) confirmed that only the `source` column contained null values, affecting close to **50% of all rows**.

Figure 4.4: Missing-value profile across all columns — only `source` is affected.

A distribution plot of the `source` column (`plotDistributions()`) further revealed that it is highly imbalanced across its non-null classes, suggesting the labelled subset may correspond to specific question domains rather than being drawn uniformly across the dataset.

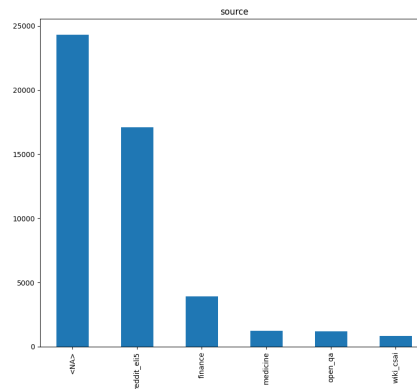


Figure 4.5: Class distribution of the **source** column, including the unlabeled (<NA>) share.

Since the missingness was isolated to a single, non-critical column, rows with a missing **source** were retained and treated as an unlabeled category rather than dropped.

4.3 Preprocessing and Feature Engineering

Handling Missing Values

Missing values in the **source** column were relabelled as an explicit **unknown** class rather than discarded, preserving the full dataset while still allowing **source** to be used as an auxiliary feature.

Label Construction

Since the raw dataset stores human and ChatGPT answers in separate columns, the data was flattened into a long format with a single unified **Y** (label) column, alongside the corresponding **X** (text) column — the standard supervised-learning shape required for classification.

id	question	source	text	label
42	What is jet lag?	reddit_eli5	Human ans A	0
42	What is jet lag?	reddit_eli5	Human ans B	0
42	What is jet lag?	reddit_eli5	ChatGPT ans	1
43	How does tax work?	finance	Human ans	0
43	How does tax work?	finance	GPT ans 1	1
43	How does tax work?	finance	GPT ans 2	1

Figure 4.6: Dataset after flattening human/ChatGPT answers into a unified text (X) and label (Y) format.

Raw-Text Feature Engineering

[41] Before embedding the text, a deliberate decision was made to first extract raw, human-interpretable statistics from it. Embeddings compress text into a geometric representation that is excellent for similarity and clustering, but which hides diagnostic signals that are useful during EDA and model debugging. The following features were engineered directly from the raw text, grouped by class (AI vs. human):

- **Word count** per answer
- **Average word length** per answer
- **Character count** per answer
- **Duplicate / near-duplicate detection**, since HC3 is known to contain repeated questions across its splits
- **Vocabulary and style statistics**, including average sentence length and punctuation frequency

These statistics were retained as standalone tabular features and later used both for standalone classical models and as auxiliary inputs alongside embeddings.

Embedding Generation

To enable correlation analysis, outlier detection, and geometric clustering, the textual data was converted into numeric embeddings using **sentence-transformers/all-MiniLM-L6-v2**. Embeddings were generated separately for the **question** column and for the **answer**

column (covering both human and ChatGPT responses), allowing the high-dimensional relationship between questions and their answers to be examined.

```
question_embedding
[ -2.27399394e-02
 -1.09591350e-01
  3.05840429e-02
  1.21594407e-03
 -4.35368456e-02
  3.48526463e-02
  5.06259054e-02  6....
```

Figure 4.7: Sample of generated `question_embedding` vectors added to the dataset.

The same procedure was repeated for the answer columns, producing the final, feature-complete version of the dataset used for modelling.

```
(zolo@zolo) ~/Desktop/Other/AbstractML
$ uv run ptixiaki.py
Running viewHead
```

index	_split	id	question	source	question_embedding	text	label	text_embedding	text_wordCount	text_avgWordLength	text_charCount
0	train	0	Why is every book I hear about a "NY Times #1 Best Seller" ? ELI5 : Why is every book I hear about a "NY Times # ...	reddit_elis	[-2.27399394e-02 -1.09591350e-01 3.05840429e-02 1.21594407e-03 -4.35368456e-02 3.48526463e-02 5.06259054e-02 6....	Basically there are many categories of "Best Seller" . Replace "Best Seller" by something like "Oscars" and eve...	0	[-8.81845430e-02 -8.28050449e-02 2.41205748e-03 -1.15142083e-02 -9.26874019e-03 9.66746733e-02 2.44084504e-02 2.7....	134	3.485074626865672	600
1	train	0	Why is every book I hear about a "NY Times #1 Best Seller" ? ELI5 : Why is every book I hear about a "NY Times # ...	reddit_elis	[-2.27399394e-02 -1.09591350e-01 3.05840429e-02 1.21594407e-03 -4.35368456e-02 3.48526463e-02 5.06259054e-02 6....	If you're hearing about it, it's because it was a very good or very well-publicized book (or both), and almos...	0	[-2.40147021e-03 -1.32534161e-01 3.24107669e-02 6.79223686e-02 1.33084469e-02 5.37671521e-02 4.61349776e-03 7....	73	3.506849315068493	328
2	train	0	Why is every book I hear about a "NY Times #1 Best Seller" ? ELI5 : Why is every book I hear about a "NY Times # ...	reddit_elis	[-2.27399394e-02 -1.09591350e-01 3.05840429e-02 1.21594407e-03 -4.35368456e-02 3.48526463e-02 5.06259054e-02 6....	One reason is lots of catagories . However, how the NY Times calculates its best seller list is n't comprehensive , ...	0	[-2.12685615e-02 -6.17095903e-02 4.98044528e-02 3.80870514e-02 -9.45346430e-03 4.19454612e-02 -6.02989048e-02	62	3.774193548387097	295

Figure 4.8: Final engineered dataset — raw text statistics, `source`, and question/answer embeddings combined.

Additional Plotting and EDA on Engineered Features

Using `PlotDataset()` from `AbstractML`, the answer embeddings [40] were projected into 3 dimensions via PCA to visually inspect class separability between AI-generated and human-generated answers.

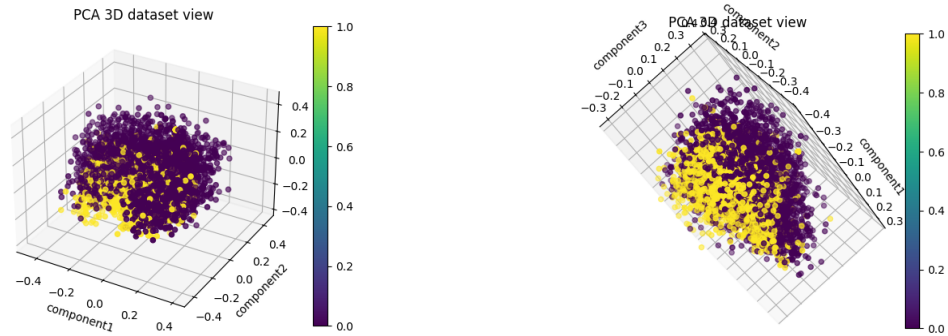


Figure 4.9: PCA projection of answer embeddings into 3D space — AI vs. human, shown from two viewing angles.

As Figure 4.9 shows, there is a visually distinguishable clustering structure between the two classes even at just 3 principal components, with the separation expected to be sharper still in the full, higher-dimensional embedding space.

A parallel 3D plot of the raw text statistics (`word count`, `char count`, `avg word length`) per class was produced for comparison.

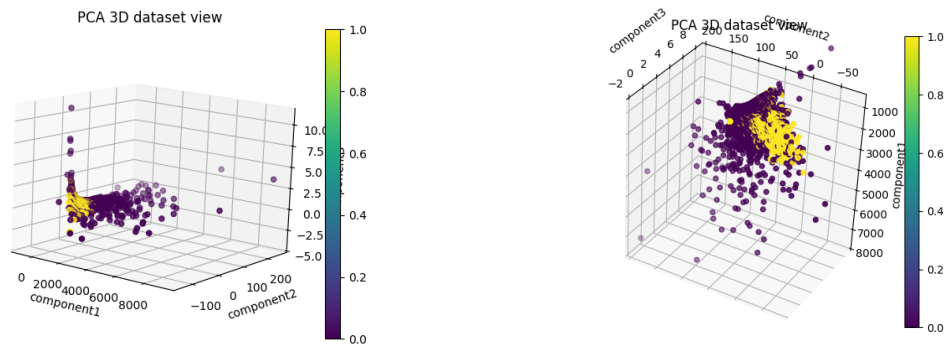


Figure 4.10: 3D scatter of raw text statistics (word count, char count, avg. word length) by class, from two viewing angles.

This shows a similarly promising, though less sharp, separation — reinforcing that both the engineered raw-text statistics and the semantic embeddings carry meaningful class signal on their own.

4.4 Train / Validation / Test Split

Following feature engineering, the final dataset consisted of **171,776 rows**, with every group of three rows corresponding to a single question (two human answers and one ChatGPT answer). The dataset was split **80/10/10** into training, validation, and test sets, with splitting performed **at the question level** rather than the row level, to prevent any leakage of the same question’s answers across splits.

Split	Proportion	Rows
Train	80%	137,626
Validation	10%	17,038
Test	10%	17,038

Table 4.1: Train / validation / test split of the processed HC3 dataset.

4.5 Initial Model Training

Given the primarily textual (NLP) nature of the dataset, a broad range of model families was evaluated — spanning classical models (feature-based and with embeddings) , and fine-tuned language models in order to compare performance and generalisation capability across paradigms.

Category	Models
Classical ML	Logistic Regression, Random Forest, Knn, Xgboost
Fine-tuned LM	SetFit (<code>sentence-transformers/all-MiniLM-L6-v2</code>)

Table 4.2: Model families and candidates evaluated for AI-text detection.

4.5.1 Validation Results

The following figures show each model’s metrics and confusion matrix on the **validation split**, before hyperparameter tuning.

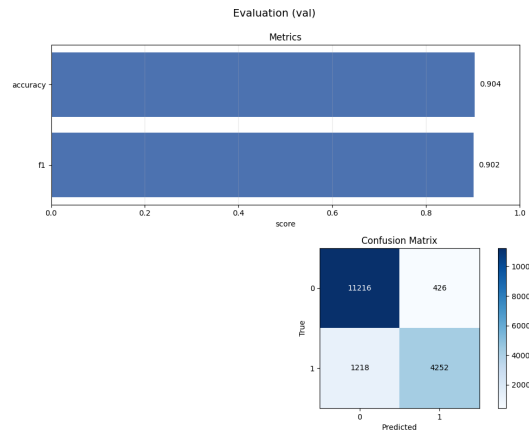


Figure 4.11: Random Forest Classifier — validation performance.

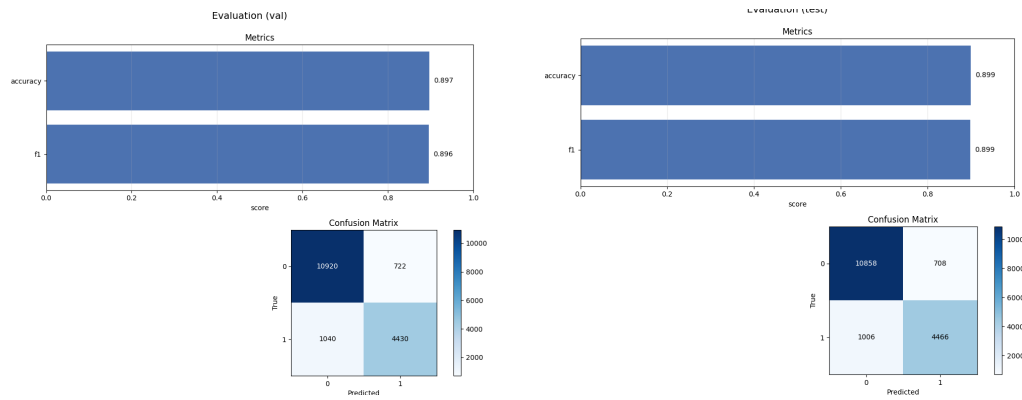


Figure 4.12: Logistic Regression Xgboost — validation performance.

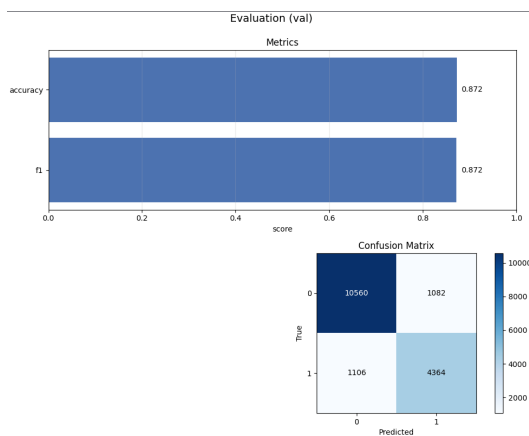


Figure 4.13: KNN Classifier — validation performance.

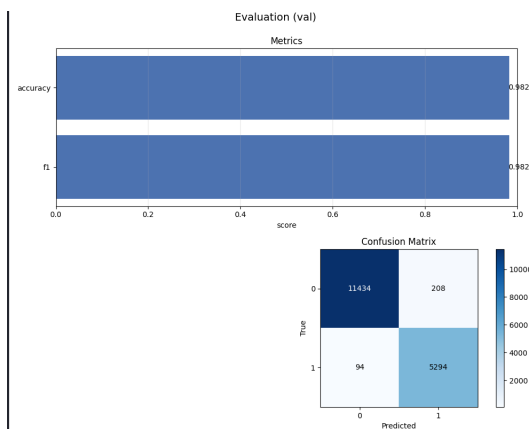


Figure 4.14: SetFit (fine-tuned MiniLM) — validation performance.

Section 4.5.2).

The SetFit fine-tuning run converged with an initial training loss profile of `{'embedding_loss': 0.4613, 'grad_norm': 1.455}` at the earliest recorded checkpoint.

4.5.2 Test Results

The following figures show each model's metrics and confusion matrix on the held-out **test split**, using the same (untuned) models evaluated above.

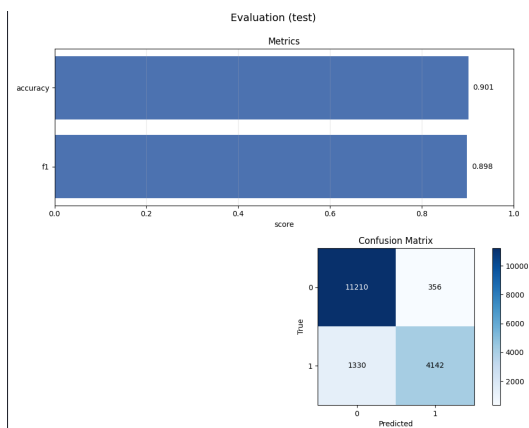


Figure 4.15: Random Forest Classifier — test performance.

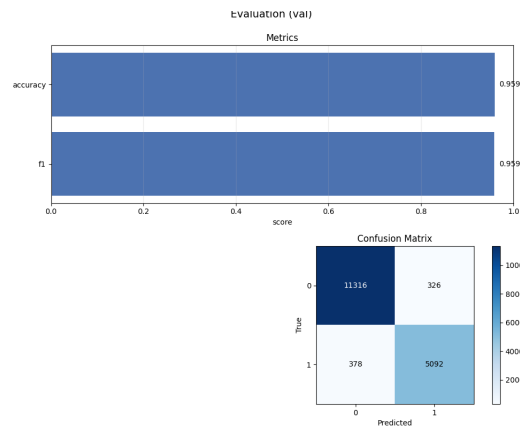


Figure 4.16: Logistic Regression — test performance.

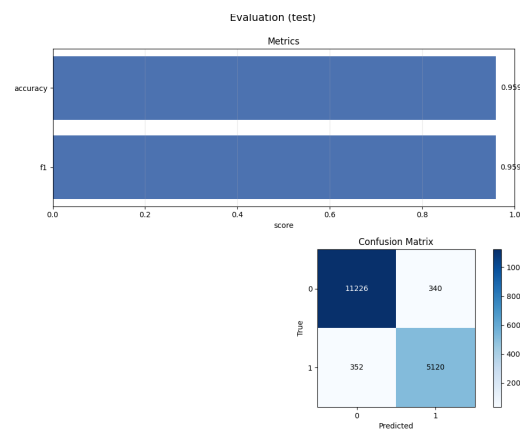


Figure 4.17: XGBoost Classifier — test performance.

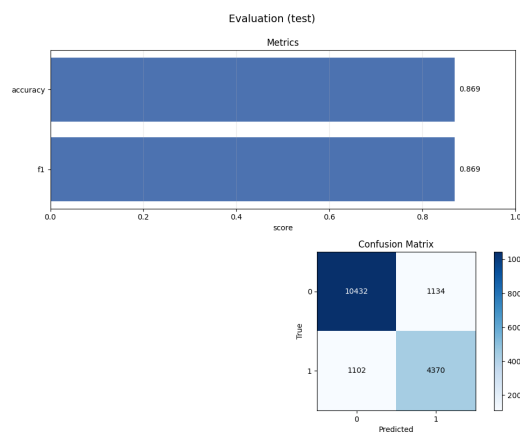


Figure 4.18: KNN Classifier — test performance.

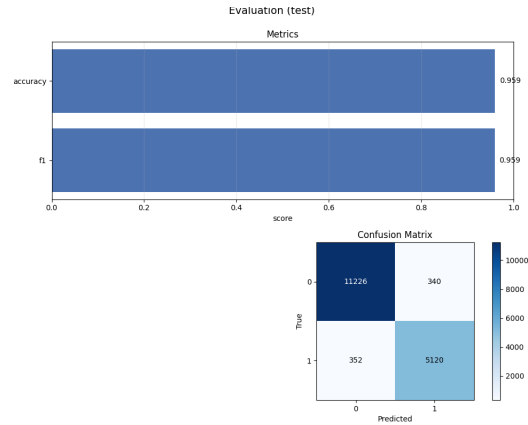


Figure 4.19: Xgboost — test performance.

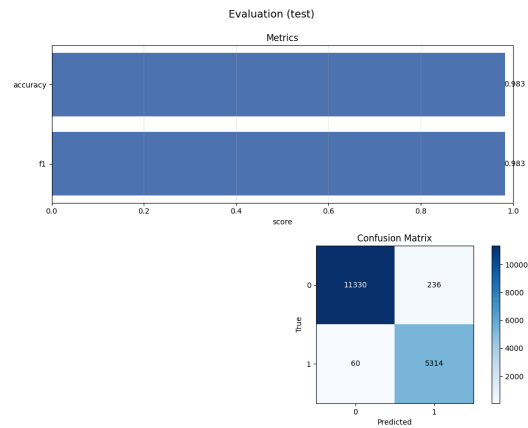


Figure 4.20: SetFit (fine-tuned MiniLM) — test performance.

4.6 Hyperparameter Optimisation (HPO)

To move from baseline models to optimised candidates, both **Random Search** and **Bayesian Search** were performed over identical, pre-defined search spaces for each classical model family, allowing a direct comparison of search-strategy efficiency in addition to raw model performance.

4.6.1 Random Search

Logistic Regression

Search space:

- C: log-uniform, range $[0.001, 10.0]$
- max_iter: $\{500, 1000, 2000\}$

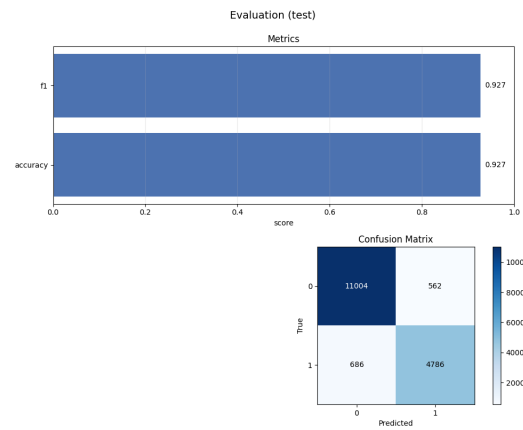


Figure 4.21: Logistic Regression — tuned metrics and confusion matrix on test data.

Best result (test set): $F1 = 0.9260$, with $C = 0.7313$, $\text{max_iter} = 2000$.

Random Forest Classifier

Search space:

- `n_estimators`: {100, 200, 300}
- `max_depth`: {10, 20, 30, None}
- `min_samples_leaf`: {1, 2, 4}

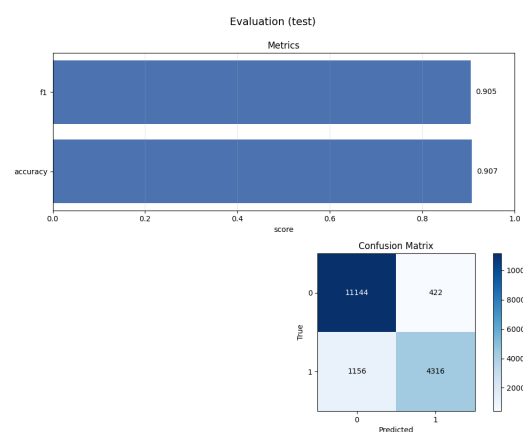


Figure 4.22: Random Forest Classifier — tuned metrics and confusion matrix on test data.

Best parameters: `n_estimators = 300`, `max_depth = 30`, `min_samples_leaf = 1`.

XGBoost Classifier

Search space:

- `n_estimators`: {100, 200, 300}
- `max_depth`: {4, 6, 8}
- `learning_rate`: log-uniform, range [0.01, 0.2]
- `subsample`: {0.8, 0.9, 1.0}

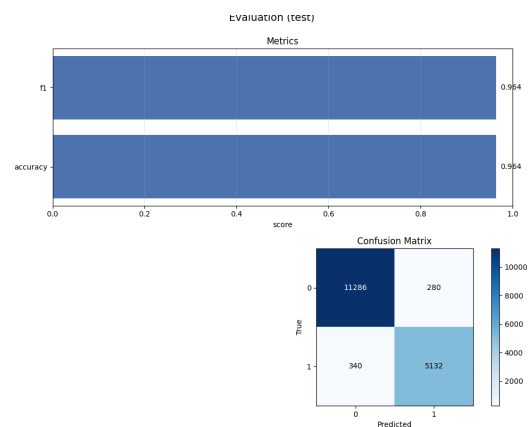


Figure 4.23: XGBoost Classifier — tuned metrics and confusion matrix on test data.

Best parameters: `n_estimators` = 300, `max_depth` = 8, `learning_rate` = 0.1448, `subsample` = 0.8.

KNN Classifier

Search space:

- `n_neighbors`: {3, 5, 7, 11, 15}
- `weights`: {uniform, distance}
- `p`: {1, 2}

Model	Best Hyperparameters
Logistic Regression	$C = 0.7313$, $\text{max_iter} = 2000$
Random Forest	$\text{n_estimators} = 300$, $\text{max_depth} = 30$, $\text{min_samples_leaf} = 1$
XGBoost	$\text{n_estimators} = 300$, $\text{max_depth} = 8$, $\text{learning_rate} = 0.1448$, $\text{subsample} = 0.8$
KNN	$[0.893]$
Setfit	$[0.983]$

Table 4.3: Summary of best hyperparameters found via search across all tuned models.

4.6.2 Bayesian Search

Logistic Regression

Same Search spaces were used for both random and Bayesian HPO.

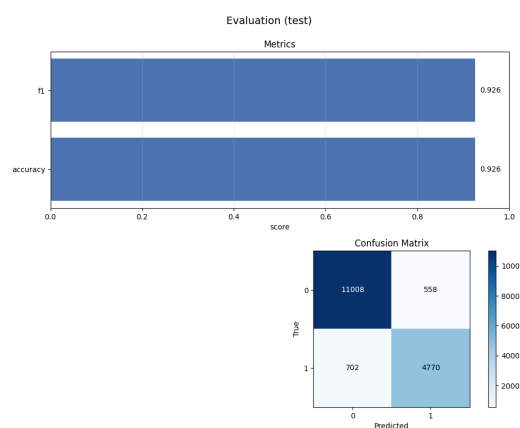


Figure 4.24: Logistic Regression — tuned metrics and confusion matrix on test data.

Random Forest Classifier

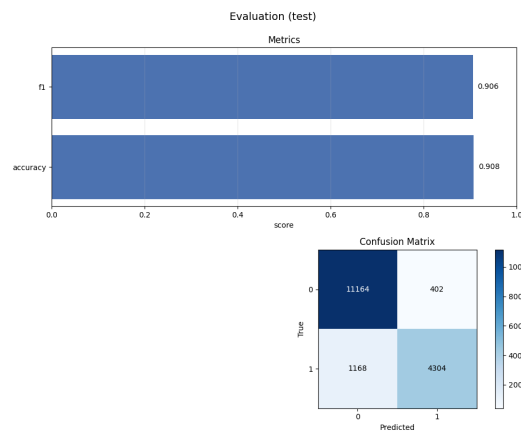


Figure 4.25: Random Forest Classifier — tuned metrics and confusion matrix on test data.

XGBoost Classifier

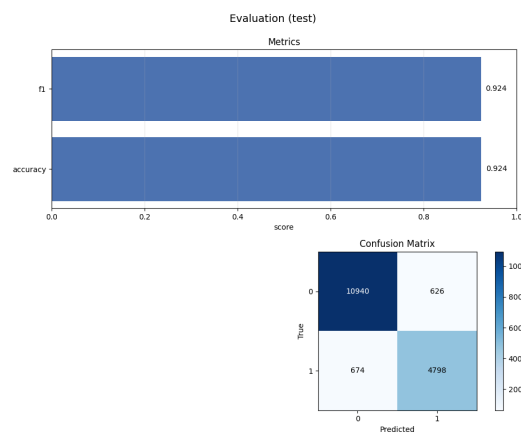


Figure 4.26: XGBoost Classifier — tuned metrics and confusion matrix on test data.

Knn Classifier

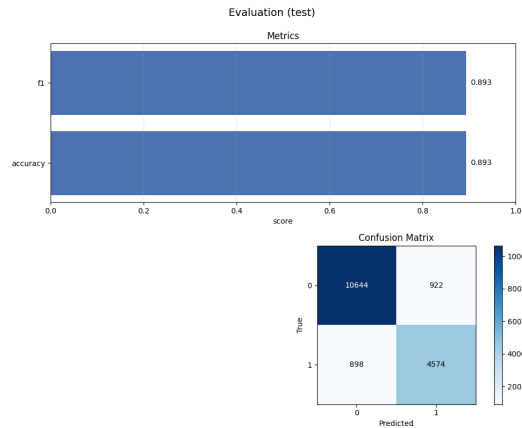


Figure 4.27: Knn Classifier — tuned metrics and confusion matrix on test data.

4.7 Final Evaluation and Statistical Significance

4.7.1 Statistical Significance

Using *AbstractML*’s funcs:

`getConfidenceInterval( )`

`testAgainstBaseline()`

`performCV()`

Across all models and both hyperparameter optimization strategies, test-set accuracy and F1 were significantly greater than the random-guess baseline of 0.5 ($\alpha = 0.05$). For example, the Bayesian-optimized Logistic Regression model achieved an accuracy of 0.9264 (95% CI [0.9224, 0.9302], Wilson interval) and an F1 score of 0.9261 (95% CI [0.9222, 0.9301], bootstrap), both significantly above baseline (binomial test, $p < 0.001$; bootstrap test, $p = 0.001$). This pattern of significance held consistently across all model–method combinations evaluated.

To assess stability without reusing the held-out test set, stratified 5-fold cross-validation was performed on `hc3_Train` using `performCrossValidation()` from *AbstractML*, with hyperparameters fixed to the best random-search configuration per sklearn model. `SetFit` was excluded due to fine-tuning cost on CPU. Table 4.4 reports mean $\pm$ standard deviation across folds.

CV scores align closely with held-out test results (e.g. XGBoost: CV F1 0.961 ± 0.002 vs. test F1 0.964) and remain well above the 0.5 random baseline, with low fold-to-fold variance ($\sigma \leq 0.005$), indicating stable generalisation rather than sensitivity to a single split.

Table 4.4: Five-fold stratified CV on `hc3_Train` (mean $\pm$ std).

Model	CV Accuracy	CV F1
Logistic Regression	0.924 ± 0.003	0.924 ± 0.003
Random Forest	0.902 ± 0.004	0.900 ± 0.004
XGBoost	0.961 ± 0.002	0.961 ± 0.002
KNN	0.888 ± 0.005	0.888 ± 0.005

4.7.2 Final Evaluation

This section summarises, side by side, the final tuned performance of every model on the held-out test set(ALL Stages- Initial Training, Random and Bayesian Search Optimization) — accuracy, precision, recall, and F1 at minimum — and identify the best-performing candidate overall. It should also report whether the difference between the top two or three models is statistically meaningful, e.g. via a paired significance test such as McNemar’s test on the test-set predictions, or bootstrapped confidence intervals on F1/accuracy, rather than relying on a single point-estimate comparison.

Table 4.5: Initial training vs. Random search vs. Bayesian optimization: best F1 / accuracy by model.

Model	Initial		Random		Bayesian	
	Best F1	Best Accuracy	Best F1	Best Accuracy	Best F1	Best Accuracy
Logistic Regression	0.959	0.959	0.927	0.927	0.926	0.926
KNN	0.869	0.869	0.893	0.893	0.893	0.893
Random Forest	0.898	0.901	0.905	0.907	0.906	0.908
XGBoost	0.959	0.959	0.964	0.964	0.924	0.924
MiniLM (SetFit)	0.983	0.983	0.980	0.980	0.98	0.98

- *Initial stage:*

Judging by all the Confusion Matrixes of all models we can clearly see that in the test- held out Dataset Evaluation the Models all performed reasonably well all over 0.86 F1 and Accuracy.(though some results could be based on luck.. thats why we performed HPO and Statistical methodologies for more robust Results).

- *Random and Bayesian Search:*

Models definitely generally Improved but not by a lot. This is due to the issue that we had a relatively small Search space alongside Search iterations(Dataset is huge for a single commercial CPU or GPU to handle). Even though that was the case we still improved with the few iterations and we would definitley improve more with bigger search space!

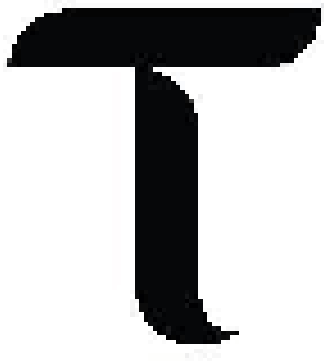
Also it must be noted that they are also Statistically Proven reliable by multiple Metrics/Conditions that we have mentioned just above (obviously on test Data).

4.8 Conclusions

We have established why AI Text Detection is gonna be important on the future and we have proposed some general Benchmark/Solutions on this specific Dataset with our own Search Space, Models and Conditions. Obviously this work (in order to be even more reliable and for models to generalize) must be trained and tested on a wider array of Datasets, Features, Train Methods, and implement Zero shot and LLM/LMs Methods for the entire stack with Stricter Learning Curve for the model to generalize no matter the model(not necessarily ChatGPT - with just this specific format of answers- different ChatGPT Models etc..) Especially if we want it to thrive into the real world and the Bittensor System where Miners are constantly Evolving and improving their Code/Models etc..

Chapter 5

Decentralized AI



*Capitalism is a **living architecture**—one that adapts, mutates, and reorganises itself around whatever technology most efficiently produces and concentrates value. Each phase of its evolution has been defined by a new answer to the same fundamental question: who controls the means of production, and how are the gains distributed? [4,5]*

From **mercantile capitalism**, where value was extracted through trade monopolies and colonial control, to **industrial capitalism**, where it was extracted through ownership of machinery and the labour of wages—to **financial capitalism**, where it was extracted through the control of credit, derivatives, and platform monopolies [4]. In every phase, the pattern is the same: a new technology emerges, a new class captures it, and the gains concentrate upward [5].

The digital economy accelerated this pattern to its logical extreme. A handful of technology companies—controlling data, distribution, and now artificial intelligence—have achieved a degree of economic concentration that mercantile monopolists could only dream of. The means of production in the twenty-first century is **intelligence**, and it is almost entirely privately owned [4,5].

What makes this transition historically significant is the nature of the commodity being produced. Every prior capitalist phase produced physical or financial goods—objects, energy, credit [4]. If intelligence becomes the dominant factor of production in the coming decades—and there is substantial evidence that it will—then the question of who owns the means of intelligence production becomes the defining political-economic question of our time. Bittensor’s answer is unambiguous: **everyone owns it, and everyone can contribute to it** [6,7].

The closest historical analogy is **open source software**. Just as Linux, Python, and GitHub allowed any developer anywhere in the world to take existing code, build upon it, and contribute improvements back to the collective—Bittensor extends this principle to intelligence itself [8]. A miner does not need to build from scratch. They can take an existing open model, stand on the shoulders of every researcher who came before them, extend it with their own ideas, and deploy it into the network.

This vision extends well beyond the academic community. NVIDIA CEO **Jensen Huang**, speaking on the future of AI, argued that while Bitcoin solved decentralised money, what the world now needs is a decentralised general-purpose intelligence layer—a commodity not of value storage, but of cognitive output.¹

¹Jensen Huang on intelligence currency: <https://youtube.com/shorts/4HodG3cX4Bk?is=rzmTdjeMwkYCa5Yh>

2

*“Bitcoin is good... but imagine a more universal currency called intelligence—
and that coin is intelligence.”*

— Jensen Huang, CEO of NVIDIA

Similarly, **Barry Silbert**, founder of Digital Currency Group, has described Bittensor as not only superior in design to Bitcoin from an incentive-alignment perspective, but as the **Linux of AI**—an open, permissionless foundation upon which the next generation of intelligent systems will be built.³

“Bittensor is the Linux of AI.”

“Bittensor is potentially a better store of value than Bitcoin.”

— Barry Silbert, Founder of Digital Currency Group

Bitcoin demonstrated that money could exist without central banks [1]. Ethereum demonstrated that contracts could execute without lawyers [2]. Bittensor demonstrates that intelligence can be produced, validated, and compensated without corporations [3,6]. This is potentially the **next phase**—one in which the means of production are owned by the network, the rules are encoded in mathematics, and the only credential that matters is the quality of your contribution [8,25].

²Jensen Huang on Templar Covenant-72B: https://youtu.be/CxQN2vM9_l0?is=ci1rPfn0a3Bu6W8

³Barry Silbert on Bittensor: <https://youtu.be/3qIL4oCe1Uo?is=L-DOwoNewmR33PSh>

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Glossary

AMM (Automated Market Maker) *A pricing mechanism that uses a mathematical formula rather than an order book to determine asset prices. Bittensor uses the constant product formula $k = \tau \times \alpha$ for subnet pools.*

Alpha Token (α) *The subnet-level token of each Bittensor subnet, hard-capped at 21 million units. Its price is determined by the AMM pool ratio $P_\alpha = \tau_{\text{pool}}/\alpha_{\text{pool}}$.*

Bonds ($B_{v,m}$) *A validator's accumulated historical commitment to a miner, updated each tempo as $B_{v,m}^{(t)} = (1 - \alpha) \cdot B_{v,m}^{(t-1)} + \alpha \cdot W_{v,m}^{(t)}$. Bonds determine a validator's dividend share and penalise weight copying.*

Consensus Score (C_m) *The final stake-weighted score assigned to a miner by Yuma Consensus: $C_m = \text{WeightedMedian}_v(W_{v,m}, \tau_v)$.*

dTAO (Dynamic TAO) *The upgrade that replaced centralised root-network emission allocation with a fully market-driven system, where each subnet's emission share is proportional to its staked τ pool.*

Emission *Newly minted τ released every block and distributed to subnet participants. Post-halving rate: 3,600 τ /day.*

Halving *A pre-programmed event that cuts the emission rate in half when a supply threshold is reached. Bittensor has two independent halving mechanisms: one for τ (network-level) and one for each subnet's Alpha token (subnet-level).*

Hotkey *A cryptographic key pair used by miners and validators to sign transactions and identify themselves on the network.*

Immunity Period *A protection window granted to newly registered subnets and miners during which they cannot be deregistered regardless of performance. For subnets, the immunity period is 4 months.*

Incentive (I_m) *A miner's final emission share: $I_m = R_m \cdot T_m$.*

Miner *A network participant who runs a computational node within a subnet, produces intelligence outputs, and earns τ proportional to their Yuma Consensus score.*

Rank (R_m) *The stake-weighted sum of all validator scores assigned to a miner:*

$$R_m = \sum_v s_v \cdot W_{v,m}.$$

Runtime *The core logic encoded into the Bittensor blockchain (Subtensor) that defines all protocol rules. Changes require governance approval.*

Staker (Delegator) *A participant who delegates τ to a validator, earning a proportional share of that validator's dividends minus a validator fee.*

Subnet *A specialised, self-contained competitive market on Bittensor dedicated to a specific computational task, governed by its own incentive mechanism and Alpha token.*

Subnet Owner *The participant who designs and maintains a subnet's incentive mechanism, receiving 18% of subnet emissions by default.*

Subtensor *The Bittensor blockchain—a Substrate-based Layer 1 that records all weights, emissions, registrations, and governance decisions.*

TAO (τ) *Bittensor's native token. Hard-capped at 21 million units. Functions as the network's reserve currency and unit of account for all economic activity.*

Tempo *One scoring round on Bittensor, consisting of 360 blocks (≈ 72 minutes). Emissions are distributed and weights are updated once per tempo.*

Trust (T_m) *A measure of how consistently validators agree on a miner's quality:*

$$T_m = \sum_v s_v \cdot \mathbb{I}[W_{v,m} > 0].$$

Validator *A participant who queries miners, evaluates their outputs, and submits weights to the blockchain. Must hold a minimum of 1,000 τ in stake to receive a validator permit.*

Validator Stake (τ_v) *The total TAO backing a validator: $\tau_v = \tau_{\text{self}} + \tau_{\text{delegated}}$.*

Weight ($W_{v,m}$) *The score a validator assigns to a miner, reflecting the perceived quality of that miner's output. Submitted to the blockchain each tempo.*

Yuma Consensus *Bittensor's incentive distribution mechanism. Aggregates validator weights into consensus miner scores and validator dividends using stake-weighted ranking, trust, and bonds.*

Block *The fundamental unit of the Bittensor blockchain, produced approximately every 12 seconds. Each block emits 0.5τ post-halving.*

Blockchain *A distributed, append-only ledger maintained by a network of independent nodes.*

Coldkey *A secure, offline key pair used to manage funds and ownership on Bittensor. Paired with a hotkey.*

Deregistration *The removal of a miner's UID from a subnet when their performance falls below the minimum threshold.*

Dividend (D_v) *The τ reward paid to validators each tempo: $D_v = \sum_m B_{v,m} \cdot I_m$.*

Proof of Work (PoW) *A consensus mechanism where participants compete to solve computationally expensive mathematical puzzles. Used by Bitcoin.*

Proof of Stake (PoS) *A consensus mechanism where validators are selected proportional to their staked capital. Used by Ethereum.*

Root Network *The top-level layer of Bittensor that receives all newly minted τ and distributes it across subnets.*

Smart Contract *Self-executing code stored on a blockchain that automatically enforces the terms of an agreement. Introduced by Ethereum.*

UID *A unique integer identifier assigned to each miner or validator registered on a subnet.*

Validator Permit *The right to submit weights and receive validator dividends on a subnet. Requires a minimum of 1,000 τ staked.*

zk-SNARKs *Zero-knowledge succinct non-interactive arguments of knowledge—cryptographic proofs that allow one party to prove knowledge without revealing information. First correctly implemented on Zcash.*

Γλωσσάριο

AMM (Αυτοματοποιημένος Διαμορφωτής Αγοράς / Automated Market Maker)

Μηχανισμός τιμολόγησης που χρησιμοποιεί έναν μαθηματικό τύπο αντί για βιβλίο εντολών (*order book*) για τον καθορισμό των τιμών των περιουσιακών στοιχείων. Το Bittensor χρησιμοποιεί τον τύπο σταθερού γινομένου $k = \tau \times \alpha$ για τα *pools* των υποδικτύων.

Token Alpha (α) Το *token* επιπέδου υποδικτύου κάθε *subnet* του Bittensor, με ανώτατο όριο προσφοράς 21 εκατομμύρια μονάδες. Η τιμή του καθορίζεται από τον λόγο του *pool* του AMM: $P_\alpha = \tau_{pool} / \alpha_{pool}$.

Δεσμοί (Bonds, $B_{v,m}$) Η συσσωρευμένη ιστορική δέσμευση ενός *validator* προς έναν *miner*, η οποία ενημερώνεται σε κάθε *tempo* ως $B_{v,m}^{(t)} = (1 - \alpha) \cdot B_{v,m}^{(t-1)} + \alpha \cdot W_{v,m}^{(t)}$. Οι δεσμοί καθορίζουν το μερίδιο μερισμάτων ενός *validator* και τιμωρούν την αντιγραφή βαρών (*weight copying*).

Βαθμολογία Συναινέσης (Consensus Score, C_m) Η τελική, σταθμισμένη ως προς το *stake*, βαθμολογία που αποδίδεται σε έναν *miner* από το Yuma Consensus: $C_m = \text{WeightedMedian}_v(W_{v,m}, \tau_v)$.

dTAO (Dynamic TAO) Η αναβάθμιση που αντικατέστησε την κεντριοποιημένη κατανομή εκπομπών μέσω του *root network* με ένα πλήρως αγοραίο σύστημα, στο οποίο το μερίδιο εκπομπής κάθε υποδικτύου είναι ανάλογο του *pool* τ που έχει γίνει *stake* σε αυτό.

Εκπομπή (Emission) Νεοκοπέντα τ που απελευθερώνονται σε κάθε *block* και διανέμονται στους συμμετέχοντες των υποδικτύων. Ρυθμός μετά το *halving*: 3.600 τ /ημέρα.

Υποδιπλασιασμός (Halving) Προκαθορισμένο γεγονός που μειώνει τον ρυθμό εκπομπής στο μισό όταν επιτευχθεί ένα συγκεκριμένο όριο προσφοράς. Το Bittensor διαθέτει δύο ανεξάρτητους μηχανισμούς *halving*: έναν για το τ (επίπεδο δικτύου) και έναν για το *token Alpha* κάθε υποδικτύου (επίπεδο υποδικτύου).

Hotkey Ζεύγος κρυπτογραφικών κλειδιών που χρησιμοποιείται από *miners* και *validators* για την υπογραφή συναλλαγών και την ταυτοποίησή τους στο δίκτυο.

Περίοδος Ασυλίας (Immunity Period) Χρονικό διάστημα προστασίας που παρέχεται σε νεοεγγεγραμμένα υποδίκτυα και *miners*, κατά το οποίο δεν μπορούν να απεγγραφούν ανεξάρτητα από την απόδοσή τους. Για τα υποδίκτυα, η περίοδος ασυλίας είναι 4 μήνες.

Κίνητρο (Incentive, I_m) Το τελικό μερίδιο εκπομπής ενός *miner*: $I_m = R_m \cdot T_m$.

Miner Συμμετέχων στο δίκτυο που λειτουργεί έναν υπολογιστικό κόμβο εντός ενός υποδικτύου, παράγει αποτελέσματα «νοημοσύνης» και κερδίζει τ ανάλογα με τη βαθμολογία του στο *Yuma Consensus*.

Κατάταξη (Rank, R_m) Το σταθμισμένο ως προς το *stake* άθροισμα όλων των βαθμολογιών που αποδίδουν οι *validators* σε έναν *miner*: $R_m = \sum_v s_v \cdot W_{v,m}$.

Runtime Η βασική λογική που είναι ενσωματωμένη στο *blockchain* του *Bittensor* (*Subtensor*) και ορίζει όλους τους κανόνες του πρωτοκόλλου. Οι αλλαγές σε αυτό απαιτούν έγκριση μέσω διακυβέρνησης (*governance*).

Staker (Delegator) Συμμετέχων που εκχωρεί τ σε έναν *validator*, λαμβάνοντας αναλογικό μερίδιο των μερισμάτων του *validator*, αφού αφαιρεθεί η προμήθεια του *validator*.

Υποδίκτυο (Subnet) Εξειδικευμένη, αυτόνομη ανταγωνιστική αγορά στο *Bittensor*, αφιερωμένη σε μια συγκεκριμένη υπολογιστική εργασία, η οποία διέπεται από τον δικό της μηχανισμό κινήτρων και το δικό της *token Alpha*.

Ιδιοκτήτης Υποδικτύου (Subnet Owner) Ο συμμετέχων που σχεδιάζει και συντηρεί τον μηχανισμό κινήτρων ενός υποδικτύου, λαμβάνοντας εξ ορισμού το 18% των εκπομπών του υποδικτύου.

Subtensor Το *blockchain* του *Bittensor* — ένα *Layer 1* βασισμένο στο *Substrate*, το οποίο καταγράφει όλα τα βάρη, τις εκπομπές, τις εγγραφές και τις αποφάσεις διακυβέρνησης.

TAO (τ) Το εγγενές *token* του *Bittensor*, με ανώτατο όριο προσφοράς 21 εκατομμύρια μονάδες. Λειτουργεί ως αποθεματικό νόμισμα του δικτύου και ως μονάδα μέτρησης για κάθε οικονομική δραστηριότητα.

Tempo Ένας γύρος βαθμολόγησης στο *Bittensor*, που αποτελείται από 360 blocks (≈ 72 λεπτά). Οι εκπομπές διανέμονται και τα βάρη ενημερώνονται μία φορά ανά tempo.

Εμπιστοσύνη (Trust, T_m) Μέτρο του πόσο σταθερά συμφωνούν οι *validators* για την ποιότητα ενός *miner*: $T_m = \sum_v s_v \cdot \mathbb{K}[W_{v,m} > 0]$.

Validator (Επικυρωτής) Συμμετέχων που υποβάλλει ερωτήματα στους *miners*, αξιολογεί τα αποτελέσματά τους και υποβάλλει βάρη στο *blockchain*. Πρέπει να διαθέτει τουλάχιστον 1.000 τ σε *stake* για να λάβει άδεια *validator*.

Stake Validator (τ_v) Το συνολικό TAO που υποστηρίζει έναν *validator*: $\tau_v = \tau_{self} + \tau_{delegated}$.

Βάρος (Weight, $W_{v,m}$) Η βαθμολογία που αποδίδει ένας *validator* σε έναν *miner*, η οποία αντανακλά την εκτιμώμενη ποιότητα των αποτελεσμάτων του *miner*. Υποβάλλεται στο *blockchain* σε κάθε tempo.

Yuma Consensus Ο μηχανισμός διανομής κινήτρων του *Bittensor*. Συγκεντρώνει τα βάρη των *validators* σε βαθμολογίες συναίνεσης για τους *miners* και σε μερίσματα για τους *validators*, χρησιμοποιώντας σταθμισμένη ως προς το *stake* κατάταξη, εμπιστοσύνη και δεσμούς.

Block Η θεμελιώδης μονάδα του *blockchain* του *Bittensor*, η οποία παράγεται περίπου κάθε 12 δευτερόλεπτα. Κάθε block εκπέμπει 0,5 τ μετά το *halving*.

Blockchain (Αλυσίδα Μπλοκ) Κατανεμημένο καθολικό (*ledger*), στο οποίο τα δεδομένα μόνο προστίθενται, και το οποίο συντηρείται από ένα δίκτυο ανεξάρτητων κόμβων.

Coldkey Ασφαλές, εκτός σύνδεσης (*offline*) ζεύγος κλειδιών που χρησιμοποιείται για τη διαχείριση κεφαλαίων και ιδιοκτησίας στο *Bittensor*. Συνδέεται με ένα *hotkey*.

Απεγγραφή (Deregistration) Η αφαίρεση του *UID* ενός *miner* από ένα υποδίκτυο όταν η απόδοσή του πέσει κάτω από το ελάχιστο όριο.

Μέρισμα (Dividend, D_v) Η ανταμοιβή σε τ που καταβάλλεται στους *validators* σε κάθε tempo: $D_v = \sum_m B_{v,m} \cdot I_m$.

Απόδειξη Εργασίας (Proof of Work, PoW) Μηχανισμός συναίνεσης στον οποίο οι συμμετέχοντες ανταγωνίζονται στην επίλυση υπολογιστικά δαπανηρών μαθηματικών γρίφων. Χρησιμοποιείται από το *Bitcoin*.

Απόδειξη Συμμετοχής (Proof of Stake, PoS) Μηχανισμός συναίνεσης στον οποίο οι *validators* επιλέγονται αναλογικά με το κεφάλαιο που έχουν δεσμεύσει (*stake*). Χρησιμοποιείται από το *Ethereum*.

Root Network Το ανώτατο επίπεδο του *Bittensor*, το οποίο λαμβάνει όλα τα νεοκοπέντα τ και τα κατανέμει στα υποδίκτυα.

Έξυπνο Συμβόλαιο (Smart Contract) Αυτοεκτελούμενος κώδικας αποθηκευμένος σε ένα *blockchain*, ο οποίος επιβάλλει αυτόματα τους όρους μιας συμφωνίας. Εισήχθη από το *Ethereum*.

UID Μοναδικό ακέραιο αναγνωριστικό που αποδίδεται σε κάθε *miner* ή *validator* που είναι εγγεγραμμένος σε ένα υποδίκτυο.

Άδεια Validator (Validator Permit) Το δικαίωμα υποβολής βαρών και λήψης μερισμάτων *validator* σε ένα υποδίκτυο. Απαιτεί τουλάχιστον 1.000 τ σε *stake*.

zk-SNARKs Συνοπτικά, μη διαδραστικά επιχειρήματα γνώσης μηδενικής γνώσης (*zero-knowledge succinct non-interactive arguments of knowledge*) — κρυπτογραφικές αποδείξεις που επιτρέπουν σε ένα μέρος να αποδείξει ότι κατέχει μια γνώση χωρίς να αποκαλύψει την ίδια την πληροφορία. Υλοποιήθηκαν σωστά για πρώτη φορά στο *Zcash*.

Declaration of AI Tool Use

In accordance with Section 5.4 of the University of Macedonia's Department's Thesis Regulations, the author declares the following use of Artificial Intelligence tools during the preparation of this thesis:

LLMs — used for [e.g. translating the glossary into Greek, debugging and formatting LaTeX code and Information gathering and collection].

All AI-assisted output was reviewed, verified and edited by the author. The research design, implementation, experiments, analysis and conclusions are the author's own work.